In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

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Abstract

Tabular foundation models (TFMs) predict labels in context without training, but they need labelled examples. We show that a frozen TFM can learn from weak labels, such as candidate sets or label proportions within bags, without updating its weights. Under exchangeability, the distribution of one unknown label given all the others is a posterior predictive with the other points as context, which is what a TFM computes. In-Context Gibbs therefore samples the latent labels with forward passes, brings in the weak labels through their likelihood, and predicts by averaging over the samples. Its exact variants target the Bayes posterior when the TFM is exact. For the sampler as run, a homogeneity assumption on the TFM gives floors and ceilings for the label accuracy of a chain, so that where a chain starts can matter. Three pre-registered studies test the method. On 28 new datasets a single chain with TabICL v2 outperformed specialised learners for label proportions (+0.052 AUC over the registered comparator) and, narrowly, for partial labels; with TabPFN v2 the same chain had fallen short under heavy ambiguity. On 20 further datasets with four TFMs, averaging chains from several starts, one of them a tuned specialised learner, improved on the single chain (+0.058, one-sided $p < 0.001$; median +0.003), and averaging over TFMs improved on a single TFM (+0.046, $p < 0.001$). Two registered criteria failed, and with one TFM the multi-start average was not significantly better than the tuned learner. In a supplementary analysis registered after Study 3 was scored, the multi-TFM average beat untuned tree-based self-training on 19 of 20 datasets and partial-label kNN on all 8; a single chain was not clearly better than the tree-based learner, and running chains for 100 instead of 30 sweeps rarely repaired a stalled chain.

1 Introduction

Labels in tabular data often arrive weak: an annotator who cannot decide records a *candidate set*, and data released under privacy constraints give only the *proportion* of each label within groups. Each form of weak supervision has its own family of learners, trained per task with a loss designed for that observation (Lv et al., 2020; Feng et al., 2020; Ardehaly & Culotta, 2017; Tsai & Lin, 2020; Chen et al., 2024b).

Tabular foundation models (TFMs) such as TabPFN (Hollmann et al., 2023; 2025) and TabICL (Qu et al., 2025; 2026) predict the label of a query point in one forward pass, conditioning on a labelled context, without training. Weak labels do not fit this interface: the context must be labelled. A TFM given one plausible label per point uses only part of what candidate sets and label proportions say (Proposition 5).

A TFM is already a Gibbs sampler for unknown labels. A TFM approximates the posterior predictive distribution (PPD) of its prior (Müller et al., 2022). Under the exchangeability built into that prior, the conditional distribution of one unknown label given all the others is precisely such a PPD, with the other points and their current labels as the context (Lemma 1). A forward pass of a frozen TFM is therefore the full conditional of a Gibbs sampler over the latent labels, and weak labels enter only through their likelihood. In-Context Gibbs samples the unknown labels block by block with forward passes, respecting candidate sets or bag counts, and predicts by averaging the TFM's predictions over the samples (Section 4). No weight of the TFM is updated; only a chain's start may use a trained model.

Evidence from three pre-registered studies. Each study fixed its hypotheses, comparators and tests before its evaluation data were touched, and each was designed around what the previous one found. Study 1, with TabPFN v2, showed large gains over the naive use of the TFM but missed its criteria: under heavy ambiguity some chains never left an uninformative labelling. Study 2 changed only the TFM, to TabICL v2, on 28 new datasets (Table 1): a single chain outperformed the specialised baselines for label proportions, by +0.052 AUC over the registered comparator and +0.041 over the strongest one, and for partial labels, by +0.012 and +0.012 accuracy over tuned CC. To study when a chain stalls we use a *response map*: the accuracy of the TFM's relabelling as a function of the accuracy of the labels in its context. Under the assumption that this map describes the TFM, it bounds where the sampler as run can go (Theorem 3); with two attractors the start matters. We therefore average chains from several starts, one of them a tuned specialised learner, and then average over TFMs. Study 3 tested both steps on 20 further datasets with four TFMs (Table 2): both hypotheses were confirmed, and the multi-TFM average matched the best TFM of each dataset without knowing which it is. Two registered criteria failed: a label-free signal did not identify failing chains across TFMs, and with one TFM the multi-start average was more than 0.02 behind the better of its single chain and the tuned learner in 14.1% of cases, against a registered limit of 10%. A supplementary analysis, registered after Study 3 was scored and run on the same splits (Section 6.5), added what reviewers of an earlier draft asked for: non-neural baselines, the naive TFM, hard-EM and an oracle for all four TFMs, and chains run for 100 sweeps. The multi-TFM average beat both non-neural baselines, the single chain did not clearly beat an untuned tree-based learner, and longer chains rarely repaired a stalled one; Section 6.6 turns these results into a recommendation.

Contributions. (1) In-Context Gibbs, a sampler over latent labels built from a frozen TFM, for instance-level and group-level weak labels, with averages over starts and over TFMs. (2) Theory: exactness with an exact TFM, stability with an approximate one, and, for the default sampler under a homogeneity assumption, invariant floors and ceilings of the label accuracy. (3) Three pre-registered studies on 72 datasets, three real partial-label datasets and a registered supplementary analysis, with the failures and the wrong predictions reported as prominently as the confirmations.

2 Related work

Weak labels. Specialised learners are trained with a loss tailored to the observation, for partial labels (Lv et al., 2020; Feng et al., 2020; Wang et al., 2022) and for label proportions (Ardehaly & Culotta, 2017; Busa-Fekete et al., 2023; Tsai & Lin, 2020; Havaldar et al., 2024); for label proportions the number of samples needed grows linearly with the bag size (Busa-Fekete et al., 2025). Unified frameworks treat many types with one EM scheme (Chen et al., 2024b;a; Shukla et al., 2023), assuming conditionally independent predictions. We replace the training of a task-specific predictor by inference with a frozen TFM: latent labels are sampled jointly, and their dependence is modelled through the TFM's context. **Tabular foundation models.** Prior-fitted networks approximate the PPD of a prior over datasets (Müller et al., 2022); later models extend their range (Grinsztajn et al., 2026) or pretrain on real tables (Ma et al., 2025), and theory studies when transformers learn PPDs in context (Kang et al., 2026). Their behaviour with imperfect or missing context labels has been studied (Hu & Ghelichi, 2026; Li et al., 2025), and iterating TabPFN on its own soft pseudo-labels has been used for semi-supervised learning (Li et al., 2025); other work pretrains new in-context models for one weak-label type (Liu et al., 2026; Fan et al., 2025; Möllers et al., 2026). We keep a general TFM frozen and condition on the weak labels through their likelihood. **Gibbs sampling with learned conditionals.** TabPFN's predictive has been used for single-pass imputation (Sepin, 2026; Feitelberg et al., 2026). Drawing labels from a PFN's predictive to resample future data underlies martingale posteriors for PFNs (Nagler & Rügamer, 2025; Ng et al., 2025), and iterating in-context learning on its own outputs samples from a model's prior (Zhu & Griffiths, 2024). Iterative imputation with chained conditional models (Liu et al., 2014) and amortised Gibbs samplers (Wu et al., 2020) are closest in spirit. Our sampler conditions every latent label on all others, uses the weak-label likelihood, and has guarantees for incompatible conditionals (Table 25, Appendix H). That iterative procedures over latent labels converge only from some starts is known for EM (Balakrishnan et al., 2014); our floors and ceilings (Theorem 3) are a counterpart for the sampler as run.

Algorithm 1 In-Context Gibbs, one chain (block version; LLP uses bag-blocks). No TFM weight is updated.

Require: features X, weak labels W with model ℓ, frozen TFM q; K blocks, T sweeps, burn-in T_0, S samples

$P_0 \leftarrow$ warm start; draw $y \sim P_0 \cdot \ell$ ▷ LLP: conditioned on bag counts
for $t = 1, \dots, T$ **do**
  partition all points (LLP: all bags) at random into blocks $B_1, \dots, B_K$
  for $k = 1, \dots, K$ **do**
    $Q \leftarrow q(\cdot \mid x_i, D_{-B_k})$ for all $i \in B_k$ ▷ one forward pass
    PLL: draw $y_i \propto Q_i \, \ell_i$ for free $i \in B_k$; LLP: draw $y_{B_g} \sim \prod_{i \in B_g} Q_i$ given counts w_g, for $B_g \subseteq B_k$
    (optional) accept the new y_{B_k} with the MH probability of Proposition 2(a)
return $\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_s q(y_* \mid x_*, (X, y^{(s)}))$ over S evenly spaced states $y^{(s)}$ with $t > T_0$

3 Problem setup

We observe features $X = (x_1, \dots, x_n)$ whose labels $y \in [C]^n$ are latent. As in the prior-fitted networks behind TFMs (Müller et al., 2022), we assume an exchangeable prior: $\phi \sim p(\phi)$ and $(x_i, y_i) \overset{\text{iid}}{\sim} p(x, y \mid \phi)$, with a test pair (x_*, y_*) from the same mechanism. For a labelled context D, the *posterior predictive distribution* (PPD) of the prior is $p(y_* \mid x_*, D)$; a TFM returns an approximation $q(y_* \mid x_*, D)$ in one forward pass. Instead of y we observe weak labels W with a known likelihood $\ell(y) = p(W \mid X, y)$, independent of (x_*, y_*) given (X, y) (A0). For *partial labels* (PLL) each point has a candidate set $S_i \ni y_i$ and $\ell(y) = \prod_i \mathbb{1}[y_i \in S_i]$, under the standard assumption that the probability of a candidate set does not depend on which member is true. For *label proportions* (LLP) the points form disjoint bags B_g with observed label counts w_g, and $\ell(y) = \prod_g \mathbb{1}[\text{counts}(y_{B_g}) = w_g]$. PU data, label noise and semi-supervised learning are further instance-level cases, $\ell(y) = \prod_i \ell_i(y_i)$ (Appendix M). The targets are the posterior over latent labels and the predictive,

$$\pi(y) = p(y \mid X, W) \propto p(y \mid X)\, \ell(y), \qquad p(y_* \mid x_*, X, W) = \textstyle\sum_y \pi'(y)\, p(y_* \mid x_*, (X, y)), \tag{1}$$

with $\pi'(y) = p(y \mid x_*, X, W)$. The analysis assumes (A1) an exact TFM, $q = p$, on every queried context; (A2) a positive PPD; and (A3) $\sum_y p(y \mid X)\ell(y) > 0$. Theorem 2 removes (A1).

4 In-Context Gibbs

Algorithm 1 gives one chain; the paragraphs below explain its steps.

The full conditional is one forward pass. Exchangeability turns a TFM into a Gibbs kernel. For every i and every assignment y with $p(y_{-i} \mid X) > 0$,

$$p(y_i \mid y_{-i}, X) = p(y_i \mid x_i, D_{-i}), \qquad D_{-i} = \{(x_j, y_j)\}_{j \neq i} \tag{2}$$

(Lemma 1): the conditional of one latent label given all the others is the PPD of a query point x_i with the rest of the current assignment as its context. Multiplying by the weak-label factor gives the full conditional of π in (1), $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\, \ell_i(k)$. A Gibbs sweep over latent labels therefore needs only forward passes of a frozen TFM, and the weak labels enter only through ℓ.

Blocks. One forward pass per label would cost n passes per sweep. We instead partition all n points at random into K blocks in every sweep and, for each block B, query the TFM once with the context D_{-B} (the other blocks with their current labels). For PLL, each free y_i, $i \in B$, is drawn from $\propto q(\cdot \mid x_i, D_{-B})\, \ell_i(\cdot)$; blocks include the points whose label is fixed, so that the context keeps its composition. For LLP, blocks are groups of whole bags, and each bag's labels are drawn from $\prod_{i \in B_g} q(y_i \mid x_i, D_{-B})$ conditioned on its counts, exactly, by dynamic programming. An optional Metropolis–Hastings (MH) step accepts or rejects the whole block, which makes the update exact (Proposition 2) at one forward pass per changed label. Contexts above 2,000 points are subsampled uniformly.

Warm start and prediction. A chain starts from a cheap estimate P_0 of the labels: for PLL, the TFM fitted on one candidate per point chosen uniformly at random, cross-fitted over 5 folds; for LLP, linear DLLP (Ardehaly & Culotta, 2017). Initial labels are drawn from P_0 restricted by ℓ. After T sweeps, of which the first T_0 are discarded, the prediction for a test point is the Rao–Blackwellised average over S evenly spaced samples,

$$\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_{s=1}^{S} q(y_* \mid x_*, (X, y^{(s)})), \tag{3}$$

This *inductive* predictor is our default; in the *transductive* variant, an ablation, test points join the chain.

Several starts and several TFMs. Section 5 shows that where a chain ends can depend on where it starts. The *multi-start* sampler runs three chains that differ only in their initial labels and averages their predictions (3) with equal weights: the *warm* start above; a *learner* start, in which P_0 is the prediction of a specialised weak-label learner L tuned on the weak labels alone (DLLP with an MLP for LLP, CC (Feng et al., 2020) for PLL); and a *cold* start drawn from the normalised likelihood. The *multi-TFM* sampler averages the multi-start predictions of several TFMs, again with equal weights. Neither average needs labels or a selection step. The TFMs stay frozen throughout, but the starts are not free of training. What is fitted, by configuration: the single chain fits nothing for PLL (its warm start uses the TFM itself) and a linear DLLP model for LLP; the multi-start and multi-TFM samplers add the learner L, tuned over 20 configurations on the weak labels; no configuration fits anything to true labels.

Cost and settings. A chain costs $TK + S$ forward passes. All settings were fixed on development datasets before any evaluation and are shared by all datasets and studies: $K = 10$, $T = 30$, $T_0 = 10$, $S = 10$ (310 passes).

5 Theory

Proofs are in Appendix A.

5.1 Exact and approximate TFMs

Lemma 1 (Full conditionals)**.** *Under the exchangeable prior, (2) holds for every i and every y with $p(y_{-i} \mid X) > 0$.*

Theorem 1 (Exactness)**.** *Assume (A0)–(A3) and $q = p$. The single-site Gibbs chain with updates $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$ for instance-level weak labels and, for LLP, the MH chain that swaps two differently labelled points of a bag are reversible with respect to π and irreducible on its support. The estimator (3) then converges almost surely to $\sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, which equals $p(y_* \mid x_*, X, W)$ for the transductive chain, and for the inductive one when the feature distribution carries no information about $p(y \mid x)$ under the prior (in general up to a TV term, Appendix A).*

With an exact TFM the exact variants of In-Context Gibbs thus compute the Bayes-optimal predictor under the TFM's prior (Proposition 1). A real TFM is only approximately a PPD, and its conditionals need not be compatible with any joint distribution. Samplers built from possibly incompatible conditionals are classical (Heckerman et al., 2000; Liu et al., 2014), and their stationary distribution depends on the scan order (Chen & Ip, 2015). The next result bounds how far it is from the posterior.

Theorem 2 (Stability under an approximate TFM)**.** *Let P be the kernel of an exact variant with the exact PPD (stationary distribution π) and $\tilde{P}$ the same kernel with a TFM $q > 0$. Then $\tilde{P}$ has a unique stationary distribution $\tilde{\pi}$, and $\mathrm{TV}(\tilde{\pi}, \pi) \le \tau(P)\, \bar{\varepsilon}$, where $\tau(P) = \sum_{t \ge 0} \delta(P^t)$ with δ Dobrushin's ergodicity coefficient and $\bar{\varepsilon} = \sum_y \tilde{\pi}(y)\, \mathrm{TV}(\tilde{P}(y, \cdot), P(y, \cdot))$.*

The bound is (integrated mixing time of the exact chain) × (error per step), a classical form (Schweitzer, 1968; Meyer, 1980; Seneta, 1988; Cho & Meyer, 2001), here non-asymptotic and with MH steps included.

Uncorrected blocks. A block update is exact when its proposal, the product of the TFM's predictives over the block, is accepted with an MH ratio (Proposition 2). The default sampler skips that step, and for it the bound above is useless: on exactly solvable problems it exceeds 1 in most cases, while the true $\mathrm{TV}(\tilde{\pi}, \pi)$ is small but does not shrink with n (Appendix A.5). Predictions behave better. In a conjugate model with unconstrained unknown labels and a fresh random block at every step, the stationary mean of the unknown-label counts equals their exact posterior mean under a prior concentration inflated by $K(n-1)/((K-1)n)$, and the resulting bias of the predictor is of order $1/((K-1)N)$ for N labelled points (Proposition 3). For the partition scan that the sampler uses this holds to leading order numerically, and for other weak labels, bag-blocks in particular, we have numerical evidence only.

5.2 The sampler as run: response maps and basins

The next result does not assume that the TFM is close to a PPD. It concerns the label accuracy of a chain of Algorithm 1 as run on instance-level weak labels: uncorrected blocks, a fresh random partition in every sweep, any deterministic TFM, fixed weak-label factors, context not subsampled. It does not cover LLP, whose bag updates are not independent draws. Let F be the set of points whose label is unknown, $n_F = |F|$, $\rho = n_F/n$, and y° the true labels, which only the analysis uses. The *label accuracy* is $r(y) = n_F^{-1} \sum_{i \in F} \mathbb{1}[y_i = y_i^\circ]$, and R_s is its value after sweep s. For a block B, $c_i(y, B)$ is the probability that the redraw of $i \in B \cap F$ is correct, and $r_{-B}(y)$ the accuracy on $F \setminus B$, the part of the context the TFM sees.

Assumption 1 (Homogeneous response, (H_δ))**.** There are a function $\Phi : [0,1] \to [0,1]$ (the *response map*), fixed offsets h_i with $|h_i| \le h_{\max}$ and $\sum_{i \in F} h_i = 0$, and $\delta \ge 0$ such that for every state y and every block B of at most $\lceil n/K \rceil$ points with $B \cap F \neq \emptyset$ and $F \not\subseteq B$,

$$\Big| \frac{1}{|B \cap F|} \sum_{i \in B \cap F} \big(c_i(y, B) - h_i\big) - \Phi\big(r_{-B}(y)\big) \Big| \le \delta.$$

(H_δ^-) keeps only the lower bound ($\ge \Phi - \delta$), (H_δ^+) only the upper bound.

The response map gives the accuracy of the TFM's relabelling of a block as a function of the accuracy of its context: it depends on *how many* context labels are right, up to per-point difficulty h_i and a tolerance δ for *which* ones are right. This is a strong, uniform requirement, which we treat as a model of the TFM to be confronted with data; Φ need not be monotone or continuous. Let $\underline{\Phi}(x) = \inf\{\Phi(r) : r \in [0,1], r \ge x\}$, $\overline{\Phi}(x) = \sup\{\Phi(r) : r \in [0,1], r \le x\}$, and for margins $\nu, \lambda > 0$ let $p(\nu, \lambda) = e^{-2n_F\lambda^2} + 3K\, e^{-c\rho n_F \nu^2}$ with $c = a^2 \min\{\frac{2}{9}, \frac{1}{18h_{\max}^2}\}$ and $a = 1 - \lceil n/K \rceil / n_F$.

Theorem 3 (Floors, ceilings and basins)**.** *Run Algorithm 1 in the setting above with $n_F > \lceil n/K \rceil$, write $p = p(\nu, \lambda)$, and let $H \ge 1$.*

- *(a)* Floor. *Under (H_δ^-), if $\underline{\Phi}(x - \nu) - \delta - \lambda \ge x$ and $R_s \ge x$, then $R_{s'} \ge x$ for all $s < s' \le s + H$ with probability at least $1 - Hp$, and the first sweep τ at which the chain is below x satisfies $\mathbb{E}[\tau - s] \ge 1/p$.*
- *(b)* Ceiling. *Under (H_δ^+), if $\overline{\Phi}(x + \nu) + \delta + \lambda \le x$ and $R_s \le x$, then $R_{s'} \le x$ for all $s < s' \le s + H$ with probability at least $1 - Hp$.*
- *(c)* Progress. *Under (H_δ^-), if $\underline{\Phi}(x - \nu) - \delta - \lambda \ge x + \gamma$ for all $x \in [r_0, b)$ and some $\gamma > 0$, then a chain with $R_0 \ge r_0$ satisfies $R_J \ge b$ for some $J \le \lceil (b - r_0)/\gamma \rceil$ with probability at least $1 - Jp$.*

For a non-decreasing Φ, read this with the *gain* $g(r) = \Phi(r) - r$. Where g exceeds a margin (set by δ, λ and the variation of Φ over ν), the chain climbs (c) and does not fall back (a); where g is below minus that margin, it does not climb (b). If g is negative on an interval and positive above it, the model has two basins: a chain that starts sufficiently far above the crossing stays up, one that starts sufficiently far below stays down, and the expected time to leave is at least $1/p$ sweeps, which for a fixed fraction of unknown labels and fixed margins grows exponentially with n_F. In this model more sweeps do not repair a start in the wrong basin, which motivates the multi-start sampler. Real chains also mix slowly: run for 100 instead of 30 sweeps (Section 6.5), 5 of 35 stalled chains reached the multi-start average or the tuned learner, while chains kept

improving on average (+0.053), most with TabPFN-3, whose label accuracy was still rising after 100 sweeps. A stalled chain is therefore a mixture of a wrong basin and slow mixing, and the theorem speaks only to the first. The constants are loose: the theorem describes a mechanism and its scaling, and at the sizes of our datasets it certifies nothing (Section 6.3, Appendix A.6).

When can the sampler help? The naive use of a TFM conditions it on one label per point drawn from the weak labels. When weak labels are generated independently of the features given the true label, the naive target ranks points as the true posterior does (Proposition 5). For PU and label noise the naive labels are the observation and nothing is lost. For PLL they carry less Fisher information than the candidate sets, and for LLP a learner that treats them as independent point labels uses about 3% of the information in the counts of bags of 8. Only for PLL and LLP, then, can the sampler recover information through ℓ, which predicted, before evaluation, where In-Context Gibbs would beat the naive TFM (Appendix M).

6 Experiments

6.1 Design: three pre-registered studies

Settings and baselines. The main claims concern LLP with random bags of 8 (ROC AUC) and PLL in which each wrong class enters the candidate set independently with probability $q \in \{0.3, 0.6\}$ (accuracy). Weak labels are generated on the training part of stratified 70/30 splits (seeds 0–2) shared by all methods; datasets have 500 to 10,000 rows. Studies 1 and 2 run the peer-reviewed baselines of each setting, in their published defaults and tuned by 20 random configurations scored on a validation split *with weak labels only*: for LLP, DLLP (Ardehaly & Culotta, 2017) with an MLP or a linear model, LLP-VAT (Tsai & Lin, 2020), EasyLLP (Busa-Fekete et al., 2023) and GLWS (Chen et al., 2024b); for PLL, PRODEN (Lv et al., 2020), GLWS/ILL (Chen et al., 2024b;a), RC and CC (Feng et al., 2020). With the same TFM we also run the *naive TFM* (one label per point consistent with the weak labels), *hard-EM* (arg max instead of sampling) and an *oracle* (the TFM given the true labels). In-Context Gibbs is never tuned per dataset. A rule applied to development datasets (set aside for choosing settings) fixed the *primary comparators*: linear DLLP in its default configuration and tuned CC. In Study 3 a *cell* is one run (TFM, dataset, seed), a *pair* a (TFM, dataset) averaged over seeds, and differences below 0.001 count as ties (Studies 1 and 2 count exact ties only).

Studies 1 and 2: one chain against specialised learners. H2 (LLP) and H4 (PLL) require the single chain to beat the primary comparator and the naive TFM (two-sided Wilcoxon signed-rank over datasets, Holm-adjusted) *and* to beat the baseline with the best mean score on $\geq$70% of datasets (LLP), or to match or beat it on $\geq$60% at each level (PLL); H3 compares it with hard-EM on log loss (Benavoli et al., 2017). *Study 1* used 24 OpenML-CC18 datasets (Bischl et al., 2021) with TabPFN v2 and, as a registered ablation, TabICL v2. *Study 2* replicated it on 28 new datasets from five curated suites (e.g. Grinsztajn et al., 2022) with TabICL v2, changing nothing else (H2′–H4′).

Study 3: several starts and several TFMs. *Study 3* used 20 further datasets (12 binary with LLP, 8 multiclass with PLL at $q = 0.6$) from the OpenML suites of Study 2, among them TabArena (Erickson et al., 2025), and OpenML-CC18, excluding every dataset used earlier and TabDPT's pretraining sets. It ran TabICL v2, TabPFN-3 (Grinsztajn et al., 2026) and TabDPT (Ma et al., 2025) on all datasets and TabPFN v2 on the 11 within its published limits. Per dataset, δ_1 is the mean over TFMs of (multi-start − single chain), and δ_2 is the multi-TFM score minus the mean over TFMs of the multi-start score, the expected score of a TFM picked at random. H1″: $\delta_1 > 0$; H2″: $\delta_2 > 0$ (one-sided Wilcoxon signed-rank, Holm-adjusted). Three criteria were registered without a test. H3″: a diagnostic computed without labels identifies the cells in which the single chain fails, i.e. is more than 0.02 below the better of the multi-start average and L (AUC $\geq$ 0.80). C4″: the multi-start sampler is more than 0.02 below the better of the single chain and the tuned learner L in at most 10% of (TFM, dataset) pairs. C5″: the multi-TFM average is on average within 0.01 of the best TFM of each dataset and more than 0.02 behind it on at most 3. The only baseline registered in Study 3 is L (tuned DLLP-MLP and tuned CC, the strongest LLP baseline and the primary PLL comparator of Study 2); Section 6.5 adds non-neural baselines. All protocols were committed

Table 1: Studies 1 and 2, pre-registered tests of the single chain. Δ: mean over datasets of (In-Context Gibbs − comparator), 3 seeds each; W/T/L: datasets won/tied/lost; p_{Holm}: two-sided Wilcoxon signed-rank, Holm-adjusted within each hypothesis. Bold: significant in favour of In-Context Gibbs. The two PLL rows against CC coincide only after rounding (Appendix D).

			Study 1: TabPFN v2			Study 2: TabICL v2, new datasets		
Setting	Metric	Gibbs vs.	Δ	W/T/L	p_{Holm}	Δ [95% CI]	W/T/L	p_{Holm}
LLP, bag size 8	AUC	DLLP (linear)	+0.024	8/0/2	0.232	+0.052 [+0.024, +0.086]	16/0/2	**<0.001**
		Naive TFM	+0.148	10/0/0	**0.004**	+0.116 [+0.085, +0.150]	18/0/0	**<0.001**
PLL, $q = 0.3$	Acc.	CC (tuned)	+0.022	10/0/4	0.084	+0.012 [+0.006, +0.019]	8/2/0	**0.016**
		Naive TFM	+0.099	13/0/1	**<0.001**	+0.022 [+0.012, +0.033]	10/0/0	**0.008**
PLL, $q = 0.6$	Acc.	CC (tuned)	−0.132	7/0/7	0.715	+0.012 [+0.006, +0.019]	8/1/1	**0.016**
		Naive TFM	+0.180	12/0/2	**0.002**	+0.205 [+0.094, +0.337]	10/0/0	**0.008**
Pre-registered verdict			H2: not met; H4: not met			H2′: met; H4′: met		

before the evaluation data were downloaded; development between studies was exploratory (Appendices B and G).

6.2 One chain with a robust TFM outperforms specialised learners

In Study 2 all pre-registered tests passed (Table 1, right). On LLP the single chain improved AUC over the primary comparator by +0.052 [+0.024, +0.086] (16/0/2) and over the strongest LLP baseline, tuned DLLP (MLP), by +0.041 [+0.018, +0.072] (better on 16/18 datasets). On PLL it improved accuracy over tuned CC by +0.012 at $q = 0.3$ and +0.012 at $q = 0.6$, a small gain close to the oracle ceiling (0.8 and 1.9 points below it), and was at least as accurate as the strongest PLL baseline on 10/10 and 9/10 datasets; at $q = 0.6$ the test does not survive a second chain (Section 6.7). With the same TFM and warm start, sampling beats taking the arg max: In-Context Gibbs had lower log loss than hard-EM in 78.9% of 38 (dataset, setting) comparisons (registered H3′) and higher mean accuracy at $q = 0.6$ (0.851 against 0.803). Averaging over samples adds little to predicting from the last sample alone ("stochastic EM": 0.843; Table 14). The gain over the naive TFM is absent where naive labels carry all the information (Appendix M).

6.3 Stalling, and what response maps do and do not explain

Study 1, with TabPFN v2, did not meet its criteria (Table 1, left). In-Context Gibbs improved on the naive TFM by +0.148 AUC on LLP and +0.099 and +0.180 accuracy on PLL, and led the primary comparators on LLP and on PLL at $q = 0.3$, short of significance. At $q = 0.6$ it beat tuned CC on 7 of the 10 datasets with fewer than eight classes (+0.033 on average), but on the four with at least eight classes chains ended far from the truth (overall −0.132): the label accuracy stayed at its starting value on mfeat-factors (0.16–0.17), reached only 0.34–0.37 on semeion and 0.44–0.64 on MiceProtein, and on optdigits one seed of three stayed low (0.27, 0.89, 0.93; Table 9). With TabICL v2 every chain on these datasets ended at 0.93 or above, the registered ablation passed every test (Appendix C), and Study 2 then tested that TFM prospectively.

Figure 1 shows the gain $\Phi(r) - r$ measured by redrawing every block given an assignment with a fraction r of correct labels and uniformly placed errors. It is positive throughout with TabICL v2 and negative for intermediate r with TabPFN v2, so the map separates the two TFMs. It is not a faithful model of the chains, however: with TabPFN v2, chains on optdigits (2 of 3 seeds) climbed through the region of negative gain. On 234 cells of development runs (both TFMs, LLP and PLL; exploratory, Appendix G), chains ended above the fixed point of the uniform-error map in 96% of cells (median error 0.073): a chain's own errors sit on points the TFM finds ambiguous and harm the context less than uniform ones. For the uniform map only the lower half of Assumption 1 is therefore plausible; it can indicate that a chain will climb, not that it will stall. With errors placed according to the chain's own marginals the fixed point is within a median of 0.016 of the final accuracy (Figure 5). Those marginals come from the chain whose end is predicted, so this shows

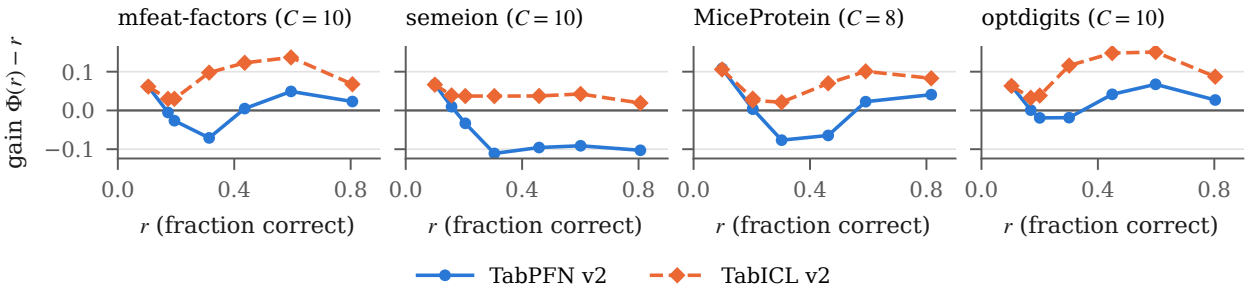


Figure 1: Response maps for PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes: gain $\Phi(r) - r$ when every block is redrawn given an assignment with a fraction r of correct labels and uniformly placed errors (needs true labels). On optdigits the context is subsampled, outside the scope of Theorem 3.

Table 2: Study 3 (20 new datasets, four TFMs, 3 seeds): registered hypotheses and criteria (Section 6.1). Differences pool AUC (LLP datasets) and accuracy (PLL datasets): mean over datasets [bootstrap 95% interval], W/T/L, one-sided Wilcoxon signed-rank p, Holm-adjusted.

	Statement	Result	Outcome
H1″	multi-start > single chain	+0.058 [+0.016, +0.114], 13/6/1, p<0.001	**confirmed**
H2″	multi-TFM > one TFM	+0.046 [+0.019, +0.079], 18/2/0, p<0.001	**confirmed**
H3″	diagnostic AUC $\geq$ 0.80	0.766 [0.59, 0.92]	not confirmed
C4″	regressions $\leq$ 10%	10/71 (14.1%)	not met
C5″	multi-TFM $\approx$ best TFM	gap 0.000; 1/20 worse by > 0.02	met

that chains settle at a fixed point of a one-dimensional map and that the placement of errors matters, not that Theorem 3 forecasts anything. Of three predictions about starts written in advance, two held (started from tuned CC, TabPFN v2 chains ended at label accuracies 0.95–0.99 on three of the datasets); one was wrong. Quantitatively the theorem is silent at these sizes: on the 3 datasets within its scope (n_F from 756 to 1,395) a floor over 30 sweeps needs a gain margin of 0.25 to 0.35 even with $\delta = 0$ (Appendix A.6), and the measured gains are at most 0.14.

6.4 Several starts and several TFMs

Confirmed. Both hypotheses of Study 3 passed (Tables 2 and 3; per dataset in Figure 4). Differences pool AUC on LLP datasets and accuracy on PLL datasets. Averaging three starts improved on the single chain by +0.058 on average (13/6/1 datasets). The gain is concentrated: +0.136 on PLL against +0.007 on LLP, a median of +0.003 over all datasets, and largest where single chains stall (TabPFN-3 on PLL: 0.371 to 0.650). The multi-TFM average improved on a TFM picked at random by +0.046 (18/2/0) and was level with the best TFM of each dataset (mean gap 0.000), more than 0.02 behind it once (micro-mass: 0.645 against 0.671). Both tests also pass without the 5 datasets flagged before the study for group or sequence structure (registered sensitivity analysis; $p_{\text{Holm}} = 0.003$ and <0.001).

Not confirmed. The diagnostic of H3″, the mean total-variation distance between the label marginals of the warm-start and learner-start chains, reached an AUC of 0.766 (66 failing cells of 213), below the registered 0.80. It separated failures for TabPFN-3 (0.93) and TabICL v2 (0.85) but not for TabDPT (0.59) or TabPFN v2 (0.64): we have no label-free diagnostic that transfers across TFMs. C4″ also failed: with one TFM, the multi-start sampler was more than 0.02 behind the better of its single chain and L in 10 of 71 pairs (14.1%): 2 behind the single chain, 9 behind L, 1 behind both (Appendix E). The registered descriptive comparison of one TFM's multi-start average with L was not significant (+0.018, $p = 0.14$).

Exploratory analysis. The following was not registered and was computed after the confirmatory scoring (Appendix E). (i) The multi-TFM average was above L on all 20 datasets, by +0.064 on average [+0.030,

Table 3: Study 3, mean test score over datasets (AUC for LLP, accuracy for PLL; number of datasets in brackets) of the single chain and the multi-start average of each TFM, of the multi-TFM average, and of the tuned learner L used in the learner start. †TabPFN v2 ran only on 10 LLP and 1 PLL datasets; its row is not comparable with the others.

	LLP, AUC (12)		PLL, acc. (8)	
TFM	single	multi	single	multi
TabICL v2	0.802	0.803	0.749	0.805
TabPFN-3	0.817	0.817	0.371	0.650
TabDPT	0.786	0.799	0.648	0.720
TabPFN v2†	0.784	0.798	0.488	0.494
Multi-TFM		**0.828**		**0.808**
Tuned learner L	0.783		0.717	

Table 4: Supplementary analysis on the 20 datasets of Study 3: mean test score over datasets (12 LLP datasets, AUC; 8 PLL datasets, accuracy). TFM-based rows average over the TFMs of each dataset. The two non-neural baselines were run in a fixed default configuration without tuning.

Method	LLP, AUC	PLL, acc.	Training
Multi-TFM average	0.828	0.808	starts use L and linear DLLP
Multi-start, mean over TFMs	0.807	0.725	starts use L and linear DLLP
Single chain, mean over TFMs	0.801	0.589	LLP start: linear DLLP
Tuned learner L	0.783	0.717	tuned on weak labels
GBDT self-training	0.796	0.519	default configuration
PL-kNN ($k = 10$)	–	0.620	none

+0.110], and by the measure of C4″ it regressed in 0 of 71 pairs; the average of single chains over TFMs, which does not use L, was above L on all 12 LLP datasets but only 6/1/1 (W/T/L) on PLL. (ii) The benefit of several starts comes from the learner start: the chain started from L alone matches the three-start average (+0.001), the cold start adds little, and averaging the single chain with L's own prediction, with no further chain, does as well (+0.008, $p = 0.26$). (iii) The three chains ended within 0.03 of each other in label accuracy in 81% of cells (91% on LLP, 61% on PLL); the PLL gain of the multi-start average is +0.328 in the other cells and +0.006 where chains agree. (iv) TabICL v2 carries much of the multi-TFM result: the average is not significantly better than TabICL v2's own multi-start average (+0.016, $p = 0.08$); averaging over TFMs buys not having to know this in advance.

6.5 Supplementary analysis registered after Study 3

Reviewers of an earlier draft asked for non-neural baselines, for the naive TFM, hard-EM and an oracle in Study 3, and for longer chains. We registered these analyses after Study 3 had been scored, together with a scoring script and one prediction, and ran them on the stored splits and weak labels of Study 3; every run was checked against the hashes of the Study 3 splits (Appendix F). They cannot change the conclusions of Study 3 and are reported as a supplement.

Non-neural baselines. We added self-training with gradient-boosted trees (pseudo-labels restricted to the candidate sets for PLL, to each bag's count for LLP) and, for PLL, a k-nearest-neighbour vote over candidate sets (PL-kNN, $k = 10$), both untuned (Table 4). The multi-TFM average beat the tree-based learner by +0.135 [+0.051, +0.234] (median +0.031, 19/0/1, $p_{\text{Holm}}<0.001$) and PL-kNN by +0.188 (8/0/0, $p_{\text{Holm}}=0.016$). The single chain did not clearly beat the tree-based learner (+0.031 [−0.003, +0.071], median +0.003, 11/1/8; descriptive): on LLP the two were level (0.801 against 0.796), and on PLL the single chain's mean was below PL-kNN (0.589 against 0.620) because of the stalled chains on datasets with many classes. The tuned learner L was not better than the better untuned non-neural baseline (+0.022, median −0.003, 7/1/12, $p_{\text{Holm}}=0.841$), so on these datasets it is a reference, not the strongest baseline.

Table 5: Score, cost and evidence of each configuration on the datasets of Study 3 (means over datasets; TFM-based rows averaged over TFMs). A pass is one forward pass of the TFM; a chain costs 310 passes, a median of 92 s to 208 s on one T4 GPU in Study 3.

Configuration	LLP, AUC	PLL, acc.	Cost per run	Trained parts	Evidence
Naive TFM	0.718	0.393	1 pass	none	registered supplement
Hard-EM with the TFM	0.800	0.539	≈310 passes	LLP start: linear DLLP	registered supplement
Single chain	0.801	0.589	310 passes	LLP start: linear DLLP	confirmatory (Studies 1–2)
Multi-start	0.807	0.725	3 chains + tuned L	tuned L	confirmatory vs. single chain
Multi-TFM average	0.828	0.808	9–12 chains + tuned L	tuned L	confirmatory vs. one TFM
GBDT self-training	0.796	0.519	CPU	default	registered supplement
PL-kNN	–	0.620	CPU	none	registered supplement
Tuned learner L	0.783	0.717	20 tuning runs	tuned	Study 3 reference

Naive TFM, hard-EM and oracle. With each of the four TFMs (Table 20 in Appendix F), the single chain improved on the naive TFM by +0.128 (18/1/1). Against hard-EM it had lower log loss on 17/0/3 datasets (two-sided $p<0.001$), as in Studies 1 and 2, but a similar score: +0.020 overall and +0.001 on LLP. The multi-TFM average stayed 0.085 below the best oracle of each dataset (0.052 on LLP, 0.135 on PLL). With TabPFN-3 on PLL the single chain (0.371) and hard-EM (0.367) stayed near the naive TFM (0.357); only the learner start moved it (0.650).

Longer chains. We reran the warm-start chain for 100 sweeps on all 72 PLL cells of TabICL v2, TabPFN-3 and TabDPT and registered one prediction (P22): fewer than a quarter of the 35 cells in which the 30-sweep chain was more than 0.02 behind the multi-start average or L would come within 0.02 of it, *and* the mean gain would be below 0.03. The first part held (5 repaired, 14%); the second did not (mean gain +0.053), so P22 was false. The gain came mostly from TabPFN-3 (+0.111), whose label accuracy rose from 0.40 to 0.49 and was still rising in many cells (Table 21).

6.6 What to run

Table 5 collects, on the datasets of Study 3, the score, the cost and the strength of evidence of each configuration. The multi-TFM average had the highest mean score in both settings, but it costs 9 or 12 chains and a tuned learner, and its advantage over L is exploratory. For label proportions a single chain, hard-EM with the same TFM and untuned tree-based self-training are level (0.801, 0.800 and 0.796 AUC); the gain over them comes from averaging over TFMs. For partial labels with many classes a single chain should not be used alone: start chains from a trained learner and average over TFMs.

6.7 Real weak labels, exactness and cost

On three real partial-label datasets (BirdSong, Lost, MSRCv2; 10 random 90/10 splits; single chain with TabICL v2, registered in Study 1), In-Context Gibbs had the highest mean accuracy on all three and the best mean rank among 10 methods (2.15; Table 32). The MH-corrected block sampler, exact when the TFM is, was not consistently better than the default at the same 30 sweeps (MH minus default with TabICL v2: +0.033 AUC on LLP, −0.021 accuracy on PLL at $q = 0.6$; Table 28) and was a median 7 times slower; the comparison confounds bias with slower mixing. Chains do not mix at the level of individual labels: in an exploratory rerun of Study 2 with a second chain, H4′ at $q = 0.6$ no longer passes ($p_{\text{Holm}} = 0.074$), while with the two chains averaged every test passes (Table 29). On one T4 GPU a chain took a median of 41 s in Study 2 and between 92 s (TabICL v2) and 208 s (TabPFN v2) in Study 3; the 9 or 12 chains of the multi-TFM average took a median of 37 minutes per dataset and seed, not counting the tuning of L.

7 Limitations

What each study supports. The comparison with specialised baselines is confirmatory only for the single chain (Studies 1 and 2), and on PLL its margin is small. Study 3 has 20 datasets (8 multiclass) and registered one baseline, the tuned learner L; its comparison with L is exploratory. The non-neural baselines

of Section 6.5 were registered after Study 3 was scored and run untuned; tuned tree-based learners might close part of the gap. The typical gain of several starts is small (median +0.003) and came from the learner start, which needs a trained learner. **Failed criteria and predictions.** We have no label-free diagnostic of stalling that transfers across TFMs; with one TFM the multi-start sampler was more than 0.02 behind the tuned learner in 9 of 71 pairs; and our prediction that longer chains would barely improve (P22) was false. **Theory versus practice.** The exactness results do not cover the uncorrected blocks used by default. Theorem 3 covers the default sampler but not LLP or subsampled contexts, rests on an assumption we could test only through its consequences along chains, with maps that need true labels, is not quantitative at our sample sizes, and does not describe slow mixing, which longer chains showed to be part of what stalls a chain. **Scope and cost.** Weak labels are synthetic except for three PLL datasets, and TFM developers may have used some of our datasets (we excluded only TabDPT's pretraining sets). The claims cover LLP with random bags of 8 and PLL with uniform distractors at $q \leq 0.6$; the sampler did not improve on the naive TFM for PU, label noise or semi-supervised learning (Appendix M). A single chain is not clearly better than untuned tree-based self-training, and the configuration that is costs up to twelve chains and a tuned learner.

8 Conclusion

Under exchangeability, a frozen TFM's predictive is the full conditional of a Gibbs sampler over unknown labels. In-Context Gibbs turns this into one procedure for candidate sets and label proportions. In pre-registered tests a single chain with a robust TFM outperformed specialised learners on new datasets, and averaging over starts and over TFMs improved on single chains; in a registered supplement the multi-TFM average also beat untuned non-neural baselines, while the single chain alone did not. Whether a chain stalls depends on the TFM and on where the chain starts, and we could neither detect stalling without labels nor reliably cure it with longer chains. Open problems: a label-free diagnostic of stalling, a cheaper way to escape a wrong start than averaging over many chains, response-map bounds with usable constants, and feature-dependent weak labels.

Broader Impact Statement

Learning from weak labels lowers the cost of annotation and makes coarse or ambiguous labels usable. Learning from label proportions also recovers individual-level predictions from data released only in aggregate; when aggregation was chosen to protect individuals, such models can weaken that protection, and releases of aggregated data should be assessed with this in mind. We have not measured how much individual-level information the sampler recovers from bag counts.

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A Proofs

Assumptions. (A0) $W \sim p(W \mid X, y)$, independent of (x_*, y_*) given (X, y); nuisance parameters of the observation model, if any, are a priori independent of (ϕ, X, y). (A1) $q(\cdot \mid x, D) = p(\cdot \mid x, D)$ for every context the algorithm queries. (A2) $p(k \mid x, D) > 0$ for all k and every context of positive probability. (A3) ℓ is known and $\sum_y p(y \mid X)\ell(y) > 0$. For instance-level observations $\ell(y) = \prod_i \ell_i(y_i)$; the set of free points is $F = \{i : |\{k : \ell_i(k) > 0\}| \geq 2\}$.

A.1 Lemma 1

Proof of Lemma 1. Given ϕ the pairs (x_j, y_j) are i.i.d., hence exchangeable, so the joint law of $(D_{-i}, (x_i, y_i))$ equals that of $(D_{n-1}, (x_*, y_*))$, a context of $n-1$ points and a query. The information (X, y_{-i}) coincides with (x_i, D_{-i}). Therefore

$$p(y_i \mid y_{-i}, X) = \frac{p((x_i, y_i), D_{-i})}{\sum_k p((x_i, k), D_{-i})} = p(y_i \mid x_i, D_{-i}). \qquad \square$$

The PPD conditions on the query features x_* as well, which may carry information about ϕ; this is the quantity the PFN training loss targets, since x_* is drawn from the prior together with the context (Müller et al., 2022).

Corollary 1 (Positivity)**.** *Under (A2), $p(y \mid X) > 0$ for every $y \in [C]^n$.*

Proof. Some y^0 has $p(y^0 \mid X) > 0$. Change one coordinate at a time, $y^0 \to y^1 \to \cdots \to y^m = y$, step k changing point s_k. If $p(y^{k-1} \mid X) > 0$, then $p(y^{k-1}_{-s_k} \mid X) > 0$, the context $D^{k-1}_{-s_k}$ has positive probability, and by Lemma 1 and (A2) $p(y^k \mid X) = p(y^{k-1}_{-s_k} \mid X)\, p(y^k_{s_k} \mid x_{s_k}, D^{k-1}_{-s_k}) > 0$. Induct on k. $\square$

A.2 Theorem 1

Proof of Theorem 1(a). The update of point i is $P_i(y, y') = \pi_i(y'_i \mid y_{-i}) \mathbb{1}[y'_{-i} = y_{-i}]$ with $\pi_i(k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$. By Lemma 1 and (A1), $\pi_i(k \mid y_{-i}) \propto p(k \mid y_{-i}, X)\ell_i(k) \propto \pi(k, y_{-i})$, the full conditional of π; hence $\pi(y)P_i(y, y') = \pi(y_{-i})\pi_i(y_i \mid y_{-i})\pi_i(y'_i \mid y_{-i})$ is symmetric in (y, y'). By Corollary 1, $\pi(y) > 0$ iff

$\ell(y) > 0$, a product set; any two of its points are joined by changing one coordinate at a time inside the set, each step with probability at least $\pi_i(\cdot \mid \cdot)/|F| > 0$, so the random-scan kernel $P = |F|^{-1} \sum_{i \in F} P_i$ is irreducible, and aperiodic since $P(y, y) > 0$. A finite, irreducible, aperiodic chain converges geometrically. The implementation sweeps the free points in a uniformly random order, $P_{\mathrm{sw}} = \frac{1}{|F|!} \sum_\sigma P_{\sigma(1)} \cdots P_{\sigma(|F|)}$; each factor preserves π, and P_{sw} is reversible because the adjoint of a product is the product in reverse order and the set of permutations is closed under reversal. □

Proof of Theorem 1(b). Choose a bag uniformly; if it is not pure, choose uniformly an ordered pair (i, j) in it with $y_i \neq y_j$ and propose y' by swapping their labels. Brook's lemma through the intermediate state (y'_i, y_{-i}) gives

$$\frac{p(y' \mid X)}{p(y \mid X)} = \frac{p(y'_i \mid y_{-i}, X)}{p(y_i \mid y_{-i}, X)} \cdot \frac{p(y'_j \mid y'_i, y_{-ij}, X)}{p(y_j \mid y'_i, y_{-ij}, X)},$$

and Lemma 1 with (A1) turns the right-hand side into the ratio R_{ij} of four TFM predictives that the algorithm computes with two forward passes. A swap preserves the counts, so $\ell(y') = \ell(y)$ and $R_{ij} = \pi(y')/\pi(y)$, independently of the order of (i, j). The number of differently labelled pairs depends only on the counts, so the proposal is symmetric and the chain is Metropolis with stationary distribution π. Two assignments of a bag with the same counts are joined by a sequence of swaps preserving the counts, each with positive probability by Corollary 1; bags are independent, so the product chain is irreducible (the lazy version is also aperiodic). □

Full statement of Theorem 1. (a) For instance-level weak labels the single-site chain satisfies detailed balance with respect to π and is irreducible and aperiodic on $\mathrm{supp}(\pi)$, hence converges geometrically. (b) For LLP the swap chain, whose acceptance ratio needs two forward passes, is reversible and irreducible on $\mathrm{supp}(\pi)$. (c) The inductive estimator (3) converges almost surely to $\bar{p}(y_* \mid x_*) = \sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, and $|\bar{p} - p(y_* \mid x_*, X, W)| \leq \mathrm{TV}\big(p(y \mid X, W),\, p(y \mid x_*, X, W)\big)$, which is zero when the prior is *separable*; the transductive chain converges to $p(y_* \mid x_*, X, W)$ exactly.

Proof of Theorem 1(c). (Inductive.) The ergodic theorem for a finite irreducible chain applied to the bounded function $y \mapsto p(y_* \mid x_*, (X, y))$, with (A1), gives almost sure convergence of (3) to $\bar{p}$. By the law of total probability and (A0), $p(y_* \mid x_*, X, W) = \sum_y p(y \mid x_*, X, W)\, p(y_* \mid x_*, (X, y))$; the difference of the expectations of one $[0, 1]$-valued function under two measures is at most their TV distance. If the prior is separable, $\phi = (\phi_x, \phi_{y|x})$ independent a priori with $p(x \mid \phi) = p(x \mid \phi_x)$ and $p(y \mid x, \phi) = p(y \mid x, \phi_{y|x})$, then $p(y \mid x_*, X, W) \propto \ell(y) \int p(y \mid X, \phi_{y|x}) p(\phi_{y|x}) d\phi_{y|x}$ times a factor that does not depend on y, so $p(y \mid x_*, X, W) = p(y \mid X, W)$ and the bound is zero. (Transductive.) Adding x_* as a free point with $\ell_* \equiv 1$ gives a chain for $p(y, y_* \mid X, x_*, W)$ that is exact by part (a) (for LLP, the swap chain combined with single-site updates of x_*, each preserving the target). By Lemma 1, the Rao–Blackwell term is $q(y_* \mid x_*, (X, y^{(s)}))$, whose average converges to $p(y_* \mid x_*, X, W)$. □

Proposition 1 (Bayes optimality under the prior)**.** *For any strictly proper scoring rule S, $p(y_* \mid x_*, X, W)$ uniquely minimises the Bayes risk $\mathbb{E}[S(f, y_*)]$ over all measurable predictors f of (X, W, x_*), the expectation being taken under the TFM's prior.*

Proof of Proposition 1. By the tower property, $\mathbb{E}[S(f, y_*)] = \mathbb{E}\big[\mathbb{E}[S(f, y_*) \mid X, W, x_*]\big]$. For each value of (X, W, x_*), f is a fixed distribution and $y_* \sim p(\cdot \mid x_*, X, W)$; strict propriety gives the unique minimiser $f = p(\cdot \mid x_*, X, W)$. Its risk is finite (at most $\log C$ for the log score). □

A.3 Theorem 2

For the random-scan single-site kernel, with $\varepsilon_{\mathrm{TV}}$ the TV error of the TFM's conditionals and κ a constant that depends only on ℓ, $\bar{\varepsilon} \leq \min\{1, 2\kappa\varepsilon_{\mathrm{TV}}\}$, and predictions inherit the error additively; both are shown at the end of the proof.

Proof of Theorem 2. Existence and uniqueness. The state space is finite, so $\tilde{P}$ has a stationary distribution. With $q > 0$ every single-site update and every MH acceptance probability is positive, so $\tilde{P}$ has the transition graph of P and is irreducible; $\tilde{\pi}$ is unique. Compatibility of the conditionals of q is never used. *Perturbation identity.* Let $\Delta = \tilde{\pi} - \pi$ and $\nu = \tilde{\pi}(\tilde{P} - P)$, so $\nu(\Omega) = 0$. From $\tilde{\pi} = \tilde{\pi}\tilde{P}$ and $\pi = \pi P$, $\Delta = \nu + \Delta P$, and iterating, $\Delta = \sum_{t<T} \nu P^t + \Delta P^T$. *Contraction.* For a signed measure μ with $\mu(\Omega) = 0$, $\|\mu K\|_1 \le \delta(K)\|\mu\|_1$. If $\tau(P) < \infty$ then $\delta(P^T) \to 0$, the remainder vanishes, and $\|\Delta\|_1 \le \|\nu\|_1 \tau(P)$; if $\tau(P) = \infty$ there is nothing to prove (no aperiodicity is needed). Finally $\|\nu\|_1 \le \sum_y \tilde{\pi}(y)\|\tilde{P}(y,\cdot) - P(y,\cdot)\|_1 = 2\bar{\varepsilon}$. *Single-site error.* $\tilde{P}(y,\cdot) - P(y,\cdot)$ averages the differences of the single-site updates, and each is the TV distance between $q\ell_i$ and $p\ell_i$ after normalisation. With $Z_p = \sum_k p(k)\ell_i(k)$, $2\,\mathrm{TV}(q_\ell, p_\ell) \le \big(\sum_k \ell_i(k)|q(k) - p(k)| + |Z_q - Z_p|\big)/Z_p \le 4(\max_k \ell_i(k)/Z_p)\,\mathrm{TV}(q,p)$, so the error is at most $2\kappa\varepsilon_{\mathrm{TV}}$ with $\kappa = \sup \max_k \ell_i(k)/Z_p$. *Prediction.* $|\mathbb{E}_{\tilde{\pi}} q - \mathbb{E}_\pi p| \le |\mathbb{E}_{\tilde{\pi}}(q-p)| + |\mathbb{E}_{\tilde{\pi}} p - \mathbb{E}_\pi p| \le \varepsilon'_{\mathrm{TV}} + \mathrm{TV}(\tilde{\pi}, \pi)$. □

Relation to classical bounds. With $E = \tilde{P} - P$ and the fundamental matrix Z, $\tilde{\pi} - \pi = \nu Z$ (Schweitzer, 1968; Meyer, 1980), which gives $\mathrm{TV}(\tilde{\pi}, \pi) \le \tau_1(Z)\bar{\varepsilon}$ with $\tau_1(Z) \le \tau(P)$ (Seneta, 1988; Cho & Meyer, 2001); the bound of Theorem 2 is the weaker, more interpretable form. It is attained by a two-state chain. For a single-site sweep the per-sweep error is at most $|F|$ times the per-update error; the MH swap kernel has error at most $4\varepsilon_\infty$, with $\varepsilon_\infty = \sup|\log q - \log p|$.

A.4 Block updates

Proposition 2 (Blocks)**.** *(a) A block update whose proposal is the product of block-wise TFM predictives (for LLP, conditioned on bag counts), accepted with the MH ratio computed by Brook's lemma along the changed labels, leaves π invariant when $q = p$. (b) Without the MH step, Theorem 2 applies with an error $\bar{\varepsilon}_B$ that is bounded only under an extra stability assumption on the PPD, and the bound grows with n for fixed K.*

Proof of Proposition 2. (a) The proposal Q_B depends only on y_{-B}, which the step does not change, so the update is MH-within-Gibbs with an independent proposal. Brook's lemma along the changed labels with Lemma 1 gives the exact ratio $p(y' \mid X)/p(y \mid X)$, multiplied by the ratio of ℓ (one for LLP). The proposal is positive on the support of the block and acceptance is positive, so the chain is irreducible. (b) Let P_{bG} be exact block Gibbs. Under the stability assumption that adding one labelled point or one unlabelled feature to a context changes the PPD by at most β in TV, the chain rule and a hybrid argument give $\mathrm{TV}\big(\prod_{i \in B \cap F} p(\cdot \mid x_i, D_{-B}),\, p(y_{B\cap F} \mid y_{-(B\cap F)}, X)\big) \le m_f(m-1)\beta$ for a block of m points of which m_f are free, and after reweighting by ℓ, $\bar{\varepsilon}_B \le 2\kappa_B\, m_f(m-1)\beta$; Theorem 2 with $P = P_{\mathrm{bG}}$ concludes. □

Why this is vacuous at our sizes. With $\beta = O(1/n)$ and fixed K ($m = n/K$), $m_f(m-1)\beta = O(n/K^2)$ grows with n. For bag-blocks, κ_B is the inverse probability that all bags of a group match their counts under the product proposal: for $n \approx 2{,}000$ and $K = 10$, $\log_{10} \kappa_B \approx 15.8$ for bags of 8. On an exactly solvable LLP problem (3 bags of 4), the uncorrected bag-block chain has $\mathrm{TV}(\tilde{\pi}, \pi)$ between 0 and about 0.6 depending on the prior and grouping, so the approximation is real at small n. Appendix A.5 explains why bounds of this kind cannot be rescued and what can be said about predictions instead.

A.5 Uncorrected blocks: bias of the predictor

Proposition 2(b) and Theorem 2 bound the distance between label distributions. For predictions a first-moment argument is sharper. We state it in a pooled conjugate model, in which it is exact.

Model. C classes with prior $\mathrm{Dir}(\alpha, \ldots, \alpha)$ and no features. Of n points, N are labelled, L_c of them in class c, and $U = n - N$ are free with no weak label (semi-supervised learning). The state is the vector $k = (k_c)$ of class counts among the free points. The exact PPD of a new point is $f_c(k) = (L_c + k_c + \alpha)/(n + C\alpha)$, so the stationary mean of the predictor (3) is the stationary mean of f_c. One step redraws the free points of a random block given all labels outside the block: the exact kernel P draws them jointly from their conditional distribution (so the posterior π is stationary), the uncorrected kernel $\tilde{P}$ independently from the PPD given the context, as Algorithm 1 does. Blocks are either *free* (a uniform subset of u free points) or *all* (a uniform

subset of $M = n/K$ of all points, the default of our sampler: the context then also loses the labelled points of the block).

Proposition 3 (Bias of the predictor under uncorrected blocks)**.** *(a) Let $P, \tilde{P}$ be Markov kernels on a finite state space and f a function with $\mathbb{E}_P[f(Y') \mid Y = y] - f(y) = A - a\, f(y)$ for all y, with $a \neq 0$. Let $d(y) = \mathbb{E}_{\tilde{P}}[f(Y') \mid y] - \mathbb{E}_P[f(Y') \mid y]$. Then for any stationary distributions π of P and $\tilde{\pi}$ of $\tilde{P}$, $\tilde{\pi}(f) - \pi(f) = \tilde{\pi}(d)/a$. (b) In the model above with free blocks, $\tilde{\pi}(f_c) = \pi(f_c)$ for all n, $u \le U$ and c. (c) With blocks of $M < n$ points over all points, $\mathbb{E}_{\tilde{\pi}}[k_c] = U(L_c + \alpha')/(N + C\alpha')$ with $\alpha' = \alpha(n-1)/(n-M)$, which is $\alpha\, K(n-1)/((K-1)n)$ for $M = n/K$: the counts of the uncorrected chain have exactly the posterior mean of the model with prior $\mathrm{Dir}(\alpha', \ldots, \alpha')$. Consequently, for $M = n/K$,*

$$\tilde{\pi}(f_c) - \pi(f_c) = \frac{U(\alpha' - \alpha)(N - CL_c)}{(n + C\alpha)(N + C\alpha')(N + C\alpha)} = \frac{\alpha(1-\lambda)(1 - Cp_c)}{(K-1)\,N} \cdot \frac{n - K}{n + C\alpha} \cdot \frac{N^2}{(N + C\alpha')(N + C\alpha)},$$

with $\lambda = N/n$ and $p_c = L_c/N$; the last two factors are $1 - O(K/n) - O(C\alpha/N)$. The bias is governed by the number of labelled points: it does not vanish as n grows with N fixed. The predictor itself is not the posterior mean under α', only the counts are.

Proof. (a) Stationarity of $\tilde{\pi}$ gives $0 = \sum_y \tilde{\pi}(y)\big(\mathbb{E}_{\tilde{P}}[f(Y') \mid y] - f(y)\big) = \tilde{\pi}(A - af + d)$, so $\tilde{\pi}(f) = (A + \tilde{\pi}(d))/a$; the same with $d \equiv 0$ gives $\pi(f) = A/a$. (b) Let j_c be the number of free points of the block currently in class c. Given the block, the context has $n - u$ points with class counts $L_c + k_c - j_c$. Under P the new labels of the block are Dirichlet–multinomial, under $\tilde{P}$ independent categorical; both have marginals $(L_c + k_c - j_c + \alpha)/(n - u + C\alpha)$, hence the same expected new count, and $d \equiv 0$ for $f = k_c$. Averaging over the block ($\mathbb{E}[j_c] = uk_c/U$) gives the drift $\mathbb{E}_P[k_c' - k_c \mid k] = A - a\, k_c$ with $a = \frac{u}{U}\big(1 - \frac{U-u}{n-u+C\alpha}\big) > 0$, so (a) applies; f_c is affine in k_c. (c) Let l be the number of labelled points in the block and $u = M - l$; l is hypergeometric with $\mathbb{E}l = MN/n$ and $\mathrm{Var}\, l = V = M\frac{N}{n}\frac{U}{n}\frac{n-M}{n-1}$. Given l, the block contains in expectation lL_c/N labelled points of class c and uk_c/U free points currently in class c, and each of its u free points is redrawn from the PPD given the context, $(L_c - l_c + k_c - j_c + \alpha)/D$ with $D = n - M + C\alpha$. Hence

$$\mathbb{E}_{\tilde{P}}[k_c' - k_c \mid k] = \frac{L_c\, \mathbb{E}[u(1 - l/N)] + \alpha\, \mathbb{E}u}{D} - k_c\, \frac{\mathbb{E}u\,(D - U) + \mathbb{E}u^2}{DU}.$$

Using $\mathbb{E}u = MU/n$, $\mathbb{E}[ul] = M^2NU/n^2 - V$ and $\mathbb{E}u^2 = V + M^2U^2/n^2$, the two brackets simplify to $\frac{MU}{n}\big[L_c\frac{n-M}{n-1} + \alpha\big]$ and $\frac{MU}{n}\big[N\frac{n-M}{n-1} + C\alpha\big]$. Setting the drift to zero under $\tilde{\pi}$ gives $\mathbb{E}_{\tilde{\pi}}k_c = U\big[L_c + \alpha\frac{n-1}{n-M}\big]/\big[N + C\alpha\frac{n-1}{n-M}\big]$, and $\frac{n-1}{n-M} = \frac{K(n-1)}{(K-1)n}$ when $M = n/K$. Under π, by exchangeability, $\mathbb{E}_\pi k_c = U(L_c + \alpha)/(N + C\alpha)$. The displayed difference follows from $f_c(k) = (L_c + k_c + \alpha)/(n + C\alpha)$ and $\alpha' - \alpha = \alpha(n - K)/((K-1)n)$. □

Reading. Part (a) is the prediction-level counterpart of Theorem 2: the bias is (expected error of one step) × (relaxation time of the mean, $1/a$), and it does not involve how far the joint distribution of a block is from the product of its marginals. In (c), removing the labelled points of the block from the context shrinks the chain towards the prior mean, by an amount proportional to $1/n$, to $1/(K-1)$, and to the ratio of free to labelled points; the bias vanishes when the labelled frequencies equal the prior mean. The closed form of (c) agrees with the fixed point of the affine drift to within 1.5×10^{-11} in 15 configurations, and the drift fixed points agree with the stationary means computed from full transition matrices to within 4×10^{-11} in 9 further configurations with $n \le 200$.

What the identity does not cover. (i) The scan of Proposition 3 draws a fresh block at every step; Algorithm 1 partitions a permutation, for which the identity is not exact: we have only numerical agreement of the leading term. (ii) With weak labels that tilt the draws (PU, PLL), the drift of the exact kernel is no longer affine and free blocks are biased: in the pooled PU model the predictor is off by up to 0.070 with free blocks and by at most 0.004 with blocks over all points, which is why the latter are our default. (iii) For LLP the uncorrected bag-block kernel is biased already in its first moment. In a model with two feature types and identical bags (33 configurations), the predictor is off by up to 0.052 when $n \le 160$ and by at most 0.0024 from $n = 320$ on ($K = 10$). (iv) A real TFM is not conjugate; the mechanism in (c) is still present, but its effective prior strength is unknown.

Why label-level bounds are vacuous. In the pooled SSL and PU models with blocks over all points (24 configurations, $n \le 500$, $K \in \{5, 10, 20\}$), the exact distance $\mathrm{TV}(\tilde{\pi}, \pi)$ lies between 0.0012 and 0.0248 and does not decrease with n (for SSL at $K = 10$: 0.005, 0.009, 0.010, 0.011 at $n = 50, 100, 200, 500$), while the bound $\tau_1(Z)\bar{\varepsilon}$ exceeds 1 in 22 of them and is 60 to 629 times too large. The predictor, in contrast, is off by at most 0.0056. A guarantee for uncorrected blocks should therefore be stated for predictions, as in Proposition 3, and not for label vectors.

A.6 Dynamics of the sampler as run

Setting. One sweep of Algorithm 1 draws a uniform permutation of $[n]$, independent of the past, and cuts it into K consecutive blocks $B_1, \dots, B_K$ of at most $m = \lceil n/K \rceil$ points. For $k = 1, \dots, K$ the free points of B_k are redrawn independently, each from the TFM's predictive given the current labels of all points outside B_k, multiplied by its weak-label factor. Every free point is redrawn exactly once per sweep. The TFM is a deterministic function of (context, query); the weak-label factors are fixed (for PU, known parameters); the context is not subsampled ($n - m$ does not exceed the context limit). We write $f_k = |B_k \cap F|$, $r_k = r_{-B_k}(Y_{s,k-1})$ for the accuracy of the context when block k is updated, $U_k = F \cap (B_1 \cup \cdots \cup B_{k-1})$ and $V_k = F \cap (B_{k+1} \cup \cdots \cup B_K)$ for the free points already and not yet updated, o_i for the correctness of point i at the start of the sweep and ξ_i for the correctness of its redraw. $\mathcal{F}_s$ is the history up to the end of sweep s. Since $n_F > m$, $F \setminus B_k \neq \emptyset$ and r_k is well defined. We use the conventions $\inf \emptyset = +\infty$ and $\sup \emptyset = -\infty$ in $\underline{\Phi}$ and $\overline{\Phi}$.

Lemma 2 (One sweep)**.** *(i) Assume* (H_δ^-) *and let* x *be an* $\mathcal{F}_s$*-measurable level with* $\underline{\Phi}(x - \nu) - \delta \ge x$*. On the event* $\{R_s \ge x\}$*,* $P\big(R_{s+1} \ge \underline{\Phi}(x - \nu) - \delta - \lambda \mid \mathcal{F}_s\big) \ge 1 - p(\nu, \lambda)$*. (ii) Assume* (H_δ^+) *and let* x *be an* $\mathcal{F}_s$*-measurable level with* $\overline{\Phi}(x + \nu) + \delta \le x$*. On* $\{R_s \le x\}$*,* $P\big(R_{s+1} \le \overline{\Phi}(x + \nu) + \delta + \lambda \mid \mathcal{F}_s\big) \ge 1 - p(\nu, \lambda)$*.*

Proof. (i) Since $|U_k| + |V_k| = n_F - f_k$,

$$(n_F - f_k)\, r_k = \sum_{i \in U_k} \xi_i + \sum_{i \in V_k} o_i. \tag{4}$$

Let $Z_k = \sum_{i \in U_k} (\xi_i - c_i)$, where c_i is the conditional probability that the redraw of i is correct given the state just before its block is updated, and let $\Xi_k = \sum_{i \in U_k} h_i + Z_k + \sum_{i \in V_k} (o_i - R_s)$ and $\mathcal{E} = \{\Xi_k > -\nu(n_F - m) \text{ for all } k\}$.

On $\mathcal{E}$*,* $r_k \ge x - \nu$ *for all* k*.* Induction on k. Suppose $r_j \ge x - \nu$ for $j < k$ (nothing to assume for $k = 1$). Then $\Phi(r_j) \ge \underline{\Phi}(x - \nu)$, and (H_δ^-) applied to each block $j < k$ with $f_j > 0$ gives $\sum_{i \in U_k} c_i \ge \sum_{i \in U_k} h_i + \sum_{j<k} f_j(\Phi(r_j) - \delta) \ge \sum_{i \in U_k} h_i + |U_k| x$. Also $\sum_{i \in V_k} o_i = |V_k| R_s + \sum_{i \in V_k} (o_i - R_s) \ge |V_k| x + \sum_{i \in V_k} (o_i - R_s)$. Adding the two in (4), $(n_F - f_k) r_k \ge (n_F - f_k) x + \Xi_k$, and on $\mathcal{E}$, since $f_k \le m$, $r_k > x - \nu(n_F - m)/(n_F - f_k) \ge x - \nu$.

End of the sweep. Every free point is redrawn once, so $R_{s+1} = n_F^{-1}\big(\sum_{i \in F} c_i + Z\big)$ with $Z = \sum_{i \in F} (\xi_i - c_i)$. On $\mathcal{E}$, by (H_δ^-), $\sum_F h_i = 0$ and $\sum_k f_k = n_F$, $\sum_{i \in F} c_i \ge \sum_k f_k(\Phi(r_k) - \delta) \ge n_F(\underline{\Phi}(x - \nu) - \delta)$, hence $R_{s+1} \ge \underline{\Phi}(x - \nu) - \delta + Z/n_F$.

Probabilities. Give each of the three parts of Ξ_k the budget $t = \nu(n_F - m)/3 = \nu a n_F/3$. The set U_k is the intersection of F with the first $(k-1)$ blocks of the permutation, which is independent of $\mathcal{F}_s$; so $\sum_{U_k} h_i$ is a sum over a sample without replacement of at most n elements of the population $\{h_i \mathbb{1}[i \in F]\}_{i \le n}$, which has sum 0 and range at most $2h_{\max}$, and by Hoeffding's inequality for sampling without replacement (Hoeffding, 1963, Section 6) it is below $-t$ with probability at most $\exp(-t^2/(2nh_{\max}^2))$. In the same way $\sum_{V_k} (o_i - R_s)$ is a sample from $\{(o_i - R_s)\mathbb{1}[i \in F]\}$, with sum 0 and range at most 1, and is below $-t$ with probability at most $\exp(-2t^2/n)$. Conditionally on the permutation, Z_k is a martingale in the order of the redraws, with at most n_F increments; each increment $\xi_i - c_i$ lies in the interval $[-c_i, 1 - c_i]$ of length 1, whose endpoints are determined by the past, so Hoeffding's lemma applied conditionally to each increment (the martingale form of Hoeffding's inequality with predictable ranges) gives $P(Z_k \le -t) \le \exp(-2t^2/n_F)$. With $n = n_F/\rho$ and $\rho \le 1$, each of the three is at most $\exp(-c\rho n_F \nu^2)$, and a union bound over $k \le K$ gives $P(\mathcal{E}^c \mid \mathcal{F}_s) \le 3Ke^{-c\rho n_F \nu^2}$. Finally $P(Z/n_F \le -\lambda) \le e^{-2n_F\lambda^2}$ by the same inequality. (If $h_{\max} = 0$ the first

part is zero and $c = 2a^2/9$.) Part (ii) is symmetric, with $\overline{\Phi}$, the upper half of the assumption and the opposite tails. □

Proof of Theorem 3. Let $\psi(x) = \underline{\Phi}(x-\nu)-\delta-\lambda$, which is non-decreasing. (a) The condition $\psi(x) \geq x$ implies $\underline{\Phi}(x-\nu)-\delta \geq x$, so by Lemma 2(i), on $\{R_{s'} \geq x\}$, $R_{s'+1} \geq \psi(x) \geq x$ with conditional probability at least $1-p$. Applying this at $s' = s, \ldots, s+H-1$ gives $P(R_{s'} \geq x \text{ for all } s < s' \leq s+H) \geq \max\{0, 1-p\}^H \geq 1-Hp$ for every integer $H \geq 1$. If $p < 1$, $\mathbb{E}[\tau - s] = \sum_{H\geq 0} P(\tau - s > H) \geq \sum_{H\geq 0}(1-p)^H = 1/p$ (τ may be infinite); if $p \geq 1$ the claim is trivial, since $\tau - s \geq 1$. (b) is symmetric, with Lemma 2(ii). (c) Let $q_0 = r_0$, $q_{j+1} = \min\{\psi(q_j), b\}$, and let J be the first index with $q_J \geq b$; since $\psi(q_j) \geq q_j + \gamma$ while $q_j < b$, $J \leq \lceil (b - r_0)/\gamma \rceil$. If $R_j \geq q_j$ and $q_j < b$, then $\underline{\Phi}(q_j - \nu) - \delta = \psi(q_j) + \lambda \geq q_j$, so Lemma 2(i) with $x = q_j$ gives $R_{j+1} \geq \psi(q_j) \geq q_{j+1}$ with conditional probability at least $1-p$. By induction $P(R_j \geq q_j \text{ for all } j \leq J) \geq 1 - Jp$. □

Remarks. (i) *What each conclusion needs.* Staying up needs only the lower half of the assumption, staying down only the upper half. The proofs use the assumption only at the pairs (state before a block update, block) of the sweeps in question, and for a floor only when the context accuracy r_{-B} is at least $x - \nu$; if a block has a single free point, the assumption constrains that point individually. (ii) *Two basins.* Suppose Φ is non-decreasing and ℓ-Lipschitz and g changes sign from negative to positive at u. Any $x > u$ with $g(x) \geq \delta + \ell\nu + \lambda$ is a floor; choosing $\nu = \lambda = (g(x) - \delta)/(\ell + 1)$ gives $p \leq (1 + 3K) \exp\big(-c\rho n_F (g(x) - \delta)^2/(\ell + 1)^2\big)$, so the expected time to fall below x grows exponentially with n_F; symmetrically below u. (iii) *Looseness.* The margin ν pays for the worst case in which all blocks of a sweep see a context that is off in the same direction, and Hoeffding's inequality uses ranges instead of variances. In simulations with a synthetic TFM whose response map is known and for which the assumption holds with $\delta = 0$ ($n = 3{,}000$, $K = 10$), the proven two-sided invariant band for $H = 60$ sweeps at total failure probability 0.05 has half-width 0.22, and the simulated 99% band 0.02. For a slope-one map with $\delta = 0$ and all labels unknown, a floor over 30 sweeps at failure probability 0.05 needs a gain margin $\nu + \lambda$ of 0.30, 0.17, 0.10 and 0.03 at $n_F = 10^3$, 3×10^3, 10^4 and 10^5. The qualitative statements hold for every n, but the bands are informative only for large n_F. (iv) *Subsampled contexts.* When the context is subsampled to a fixed size, the accuracy the TFM sees is that of a random subsample, which adds a deviation term to ν; we do not state that extension, and the empirical check of Section 6.3 includes cells with subsampled contexts. (v) *LLP.* Bag updates draw the labels of a bag jointly given its counts, so the redraws within a block are not independent and the lemma does not apply.

Proposition 4 (Stationary moments for an affine map)**.** *In the setting of Lemma 2, let all points be free ($\rho = 1$), K divide n, and suppose that every redraw in block k is correct, independently given the context, with probability exactly $c_0 + b\, r_k$, where $0 < c_0$, $c_0 + b < 1$ and $0 \leq b < 1$ (a point-wise affine response, stronger than Assumption 1 with $\Phi(r) = c_0 + br$, $h \equiv 0$, $\delta = 0$, which constrains block averages only). Then (a) $\mathbb{E}[R_{s+1} \mid \mathcal{F}_s] = \alpha_0 + \alpha_1 R_s$ and $\operatorname{Var}[R_{s+1} \mid \mathcal{F}_s] = q_0 + q_1 R_s + q_2 R_s^2$, with coefficients given by a K-step recursion; (b) under the stationary distribution of the chain, $\mathbb{E}R = r^* = c_0/(1-b)$, the lag-one autocorrelation of (R_s) is α_1, $|\alpha_1| < 1$, and $\operatorname{Var} R \cdot (1 - \alpha_1^2 - q_2) = q_0 + q_1 r^* + q_2 r^{*2}$.*

Proof. Let $S = nR_s$, O_j the number of correct labels in block j at the start of the sweep, and N_k the number of correct redraws in block k. Because the partition is uniform, $(O_1, \ldots, O_K)$ is multivariate hypergeometric: its mean is affine in S and its covariance matrix is quadratic in S. Since $r_k = \big(\sum_{j<k} N_j + \sum_{j>k} O_j\big)/(n-m)$, we have $N_k = mc_0 + g_k^\top W_k + \varepsilon_k$, where $W_k = (N_{<k}, O)$, g_k is a constant vector and $\varepsilon_k = N_k - m(c_0 + br_k)$ has conditional mean zero given the past of the sweep and conditional variance $m(c_0 + br_k)(1 - c_0 - br_k)$, a quadratic function of W_k whose expectation needs only the mean and covariance of W_k. By induction on k, the mean of $(N_{\leq k}, O)$ is affine in S and its covariance quadratic in S; $R_{s+1} = \sum_k N_k/n$ inherits both, which is (a). For (b), the chain is finite, irreducible and aperiodic because every redraw is correct with probability in $(0, 1)$, so it has a unique stationary distribution, under which $\mathbb{E}R = \alpha_0 + \alpha_1 \mathbb{E}R$. If $\alpha_1 = 1$ then $\alpha_0 = 0$ and $\mathbb{E}[R_{s+1} \mid R_s = 0] = 0$, impossible since $c_0 > 0$; so $\mathbb{E}R = \alpha_0/(1 - \alpha_1)$. The recursion for the means is affine in S and can be evaluated at any real S; at $S = nr^*$ it gives $\mathbb{E}r_k = r^*$ and $\mathbb{E}N_k = m(c_0 + br^*) = mr^*$ for every k, so r^* is the fixed point of $R \mapsto \alpha_0 + \alpha_1 R$. The covariance of R_{s+1} and R_s is $\alpha_1 \operatorname{Var} R$. The law of total variance gives $\operatorname{Var} R = \mathbb{E}[\operatorname{Var}(R_{s+1} \mid R_s)] + \alpha_1^2 \operatorname{Var} R$; the first term is positive, so $\alpha_1^2 < 1$, and expanding it with $\mathbb{E}R^2 = \operatorname{Var} R + r^{*2}$ gives the last identity. □

Table 6: Proposition 4 against simulation with the actual sampler and a synthetic TFM with a known affine response map (PLL, $C = 10$, $q = 0.6$, $K = 10$; per-point offsets with $h_{\max} = 0.02$, which the proposition ignores). "Jacobi": the approximation that treats all blocks of a sweep as seeing the same context.

$\Phi(r)$	n	s.d. (exact)	s.d. (simulated)	ratio	s.d. (Jacobi)	99% band (Gaussian)	99% band (simulated)
$0.3 + 0.5r$	500	0.0346	0.0341	0.985	0.0274	[0.511, 0.689]	[0.512, 0.686]
$0.3 + 0.5r$	3,000	0.0142	0.0143	1.011	0.0112	[0.564, 0.636]	[0.563, 0.636]
$0.45 + 0.5r$	500	0.0190	0.0191	1.005	0.0155	[0.851, 0.949]	[0.844, 0.944]
$0.45 + 0.5r$	3,000	0.0078	0.0077	0.996	0.0063	[0.880, 0.920]	[0.880, 0.919]

The exact standard deviation agrees with the simulated one within a factor 0.985–1.011 (Table 6); the Jacobi approximation is 18%–21% too small, because the noise of early blocks propagates to later blocks through the context. The Gaussian band is an approximation; the proposition gives moments, not tails.

A.7 When the naive TFM suffices

Let $\tilde{\eta}_k(x) = P(\tilde{w} = k \mid x)$ be the function that a consistent learner fitted on naive labels $\tilde{w}$ approaches, and $\eta_k(x) = p(y = k \mid x)$ the true posterior.

Proposition 5 (When the naive TFM suffices)**.** *(a) If the naive label depends on x only through y, with equal off-diagonal entries in each column of the channel, then $\tilde{\eta}_k = b_k + \lambda_k \eta_k$ with $\lambda_k > 0$, so the naive target ranks points exactly as η_k; this covers PU-SCAR, class-dependent noise, SSL with labels missing completely at random, PLL with independent distractors and LLP with random bags. (b) If the naive labels are drawn from the observation by a kernel that does not depend on the parameter, their Fisher information is at most that of the observation, with equality iff the observation's score is a function of the naive labels. Equality holds for PU-SCAR and label noise, where the naive labels are the observation, and for LLP when the naive labels match the bag counts. For PLL the inequality was strict in every case we computed. For LLP the loss arises at the learner: the point-wise (composite) likelihood on naive labels has a Godambe information of about 3% of the information in the counts for bags of 8 in a logistic model.*

Complement to Proposition 5(a). For binary labels with class prior ϖ, for any score s, $\mathrm{AUC}(\eta) - \mathrm{AUC}(s) \le \mathbb{E}|s - \tilde{\eta}|/(\lambda\varpi(1-\varpi))$: the AUC regret of a learner fitted to naive labels is its own estimation error, amplified by $1/\lambda$.

Proof of Proposition 5 and of the complement. (a) With $P(\tilde{w} = k \mid x, y) = M_{y,k}$ independent of x and $M_{y,k} = b_k$ for $y \neq k$, $\tilde{\eta}_k = M_{k,k}\eta_k + b_k(1-\eta_k) = b_k + \lambda_k\eta_k$, $\lambda_k = M_{k,k} - b_k$. For PLL with each wrong class entering the candidate set independently with probability q and the naive label chosen uniformly from the set, $M_{y,y} = a = \mathbb{E}[1/(1+N)] = \frac{1-(1-q)^C}{Cq}$ with $N \sim \mathrm{Bin}(C-1, q)$ and $b = (1-a)/(C-1)$, so $\lambda = (Ca-1)/(C-1) > 0$ for $q < 1$. For LLP with random bags of size k, $P(\tilde{w}_i = 1 \mid x_i, y_i) = (y_i + (k-1)\varpi)/k$, so $\lambda = 1/k$. A strictly increasing affine map preserves ranks, hence one-vs-rest AUC. For the binary bound, write $\varpi(1-\varpi)\mathrm{AUC}(s) = \mathbb{E}[\eta(X)(1-\eta(X'))h_s(X, X')]$ for independent copies, with $h_s = \mathbb{1}[s(X) > s(X')] + \frac{1}{2}\mathbb{1}[s(X) = s(X')]$; symmetrising, $\varpi(1-\varpi)[\mathrm{AUC}(\eta) - \mathrm{AUC}(s)] = \frac{1}{2}\mathbb{E}\big[|\Delta|(\mathbb{1}[s \text{ reverses the pair}] + \frac{1}{2}\mathbb{1}[s \text{ ties}, \Delta \neq 0])\big]$ with $\Delta = \eta(X) - \eta(X')$. With $t = (s-b)/\lambda$, which has the ranks of s, a reversed or tied pair has $|\Delta| \le |\eta(X) - t(X)| + |\eta(X') - t(X')|$, so the right-hand side is at most $\mathbb{E}|s - \tilde{\eta}|/\lambda$. For $\lambda = 1$ and scores that are class-probability estimates, this is Corollary 12 of Agarwal (2014). (b) $p_\theta(\tilde{w}) = \sum_O p_\theta(O) R(\tilde{w} \mid O)$ gives $\partial_\theta \log p_\theta(\tilde{w}) = \mathbb{E}_\theta[\partial_\theta \log p_\theta(O) \mid \tilde{w}]$, and Jensen's inequality for conditional expectations gives $I_{\tilde{w}}(\theta) \le I_O(\theta)$, with equality iff the score of O is $\sigma(\tilde{w})$-measurable. □

Information lost by the naive labels (logistic model $\eta = \sigma(\theta x)$, $\theta = 2$, $x \sim \mathcal{N}(0, I)$; per point). LLP: the Godambe information of the pointwise (composite) likelihood on naive labels, relative to the information of the bag counts, is 0.134, 0.060, 0.033, 0.017, 0.009 for bags of 2, 4, 8, 16, 32. PLL: the naive labels retain 0.24–0.26 of the information in the candidate sets at $q = 0.3$ and 0.06–0.10 at $q = 0.6$ ($C \in \{3, 4, 7\}$). For PU-SCAR and label noise the ratio is 1.

B Pre-registration and timeline

All three studies were run under a written protocol with hypotheses, comparators, tests, datasets and decision rules fixed before any evaluation data were used; every later change was logged as a dated deviation. In the anonymised repository each freeze is a commit, and every result file records the hashes of the code that produced it.

Table 7: Timeline of the three studies (2026, UTC). "Exploratory" marks analyses that were not part of the registered tests.

Date	Event
26 Oct	Study 1 protocol frozen (hypotheses H1–H5, datasets, baselines, tuning spaces, primary-comparator rule, tests, ablations, sensitivity analysis). Later changes were logged as dated deviations: all but three before the evaluation runs; the three made during the analysis (dropping an external LLP benchmark whose data were unavailable, fixing the reading of an ambiguous rule to its weakest form, labelling the TFM ablation exploratory) are reported.
28 Oct–1 Nov	Study 1 runs on the evaluation datasets (TabPFN v2; registered TabICL v2 ablation; baselines).
30 Oct	Study 1 analysis: H2 and H4 not met, H3 and H5 met; TFM ablation passes all tests (exploratory). Decision to run a confirmatory replication changing only the TFM.
1 Nov	Study 2 dataset census and selection from metadata only; flags for the sensitivity analysis set by hand.
1 Nov 15:19	**Study 2 frozen** (commit), before any Study 2 table was downloaded.
1–2 Nov	Data download and post-check rule applied (two datasets fail $n \geq 500$ after de-duplication; first reserve added, subsampled to 10,000 rows). Exploratory response-map runs on Study 1 datasets only.
2–3 Nov	Study 2 runs; confirmatory analysis with the pre-written script. One post-hoc fix of a floating-point tie in the analysis (a tie counted as a loss); no conclusion changed. Failed cells of one baseline rerun once.
3 Nov	Exploratory runs on Study 1 datasets: $\widehat{R}$ and MH-corrected blocks with TabICL v2.
4–10 Nov	Development of the multi-start and multi-TFM samplers on earlier and development datasets (exploratory; Appendix G); analysis of the response-map dynamics (Theorem 3).
10 Nov 04:21	**Study 3 frozen** (commit), before its code, its scoring script and any download of its tables.
10–12 Nov	Study 3 runs (three sessions); one interim look at 60 of 71 pairs on 11 Nov; confirmatory scoring with the pre-written script on 12 Nov; exploratory decomposition afterwards (Appendix E).

C Study 1 with TabICL v2 (registered TFM ablation)

Table 8 applies the H2 and H4 tests of Study 1 to the registered TFM ablation, in which every TFM-based method uses TabICL v2. With clean labels the two TFMs are about equally accurate (oracle within 0.005); the difference appears only with weak labels. Because TabICL v2 was promoted after these results were seen, they are exploratory; Study 2 is the confirmatory test.

Table 8: Study 1 datasets, TabICL v2 for all TFM-based methods (exploratory). Same tests and comparators as Table 1.

Setting	Comparator	Δ [95% CI]	W/T/L	p_{Holm}	mean: In-Context Gibbs (TabICL / TabPFN)		/ oracle (TabICL)
LLP, bag 8	DLLP (linear)	+0.058 [+0.029, +0.093]	10/0/0	0.004	0.889	0.855	0.928
PLL, $q = 0.3$	CC, tuned	+0.029 [+0.013, +0.051]	12/0/2	0.004	0.854	0.847	0.865
PLL, $q = 0.6$	CC, tuned	+0.038 [+0.013, +0.071]	12/0/2	0.009	0.831	0.661	0.865

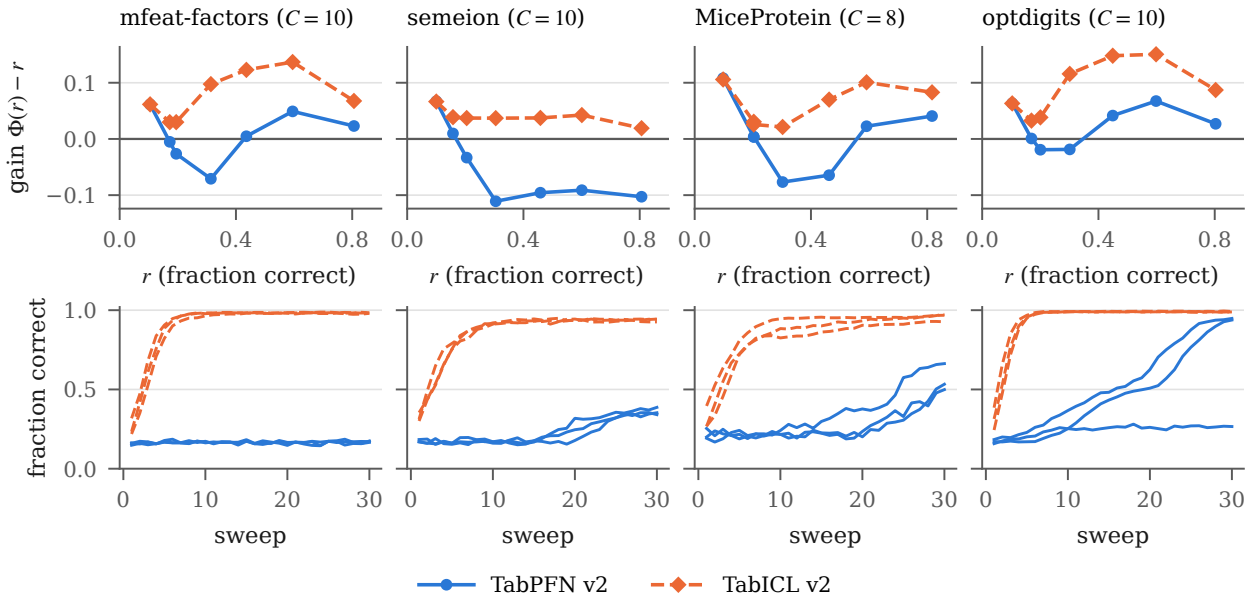


Figure 2: PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes. Top: response maps as in Figure 1. Bottom: fraction of correct training labels along the chains of the main runs, one line per seed (diagnostic only; the sampler never sees the true labels). With TabPFN v2 several chains end far below the truth, though not all stay within the region of negative gain (Table 9); with TabICL v2 all chains reach high label accuracy within 5–15 sweeps.

Table 9: Label accuracy of the single chains of Study 1 (PLL, $q = 0.6$) on the four datasets with at least eight classes: mean value after the first sweep, final value (mean of the last five sweeps) and test accuracy for each of the three seeds, sorted. Label accuracies use the true training labels and are diagnostics only.

TFM	Dataset	after sweep 1	final label accuracy (3 seeds)	test accuracy (3 seeds)
TabPFN v2	mfeat-factors	0.16	0.16, 0.17, 0.17	0.10, 0.10, 0.10
TabPFN v2	semeion	0.17	0.34, 0.35, 0.37	0.27, 0.30, 0.31
TabPFN v2	MiceProtein	0.22	0.44, 0.44, 0.64	0.43, 0.44, 0.56
TabPFN v2	optdigits	0.17	0.27, 0.89, 0.93	0.29, 0.92, 0.96
TabICL v2	mfeat-factors	0.26	0.98, 0.98, 0.98	0.97, 0.98, 0.99
TabICL v2	semeion	0.32	0.93, 0.94, 0.94	0.95, 0.95, 0.96
TabICL v2	MiceProtein	0.31	0.93, 0.96, 0.96	0.92, 0.93, 0.96
TabICL v2	optdigits	0.31	0.99, 0.99, 0.99	0.99, 0.99, 1.00

D Study 2: sensitivity analysis, all baselines and per-dataset results

Sensitivity. Five datasets were flagged before any data were downloaded (repeated measurements of the same units, or temporal drift): SpeedDating and eye_movements (binary); JapaneseVowels, cjs and gas-drift (multiclass). Without them, every conclusion is unchanged (Table 10). With seven multiclass datasets, Holm-adjusted Wilcoxon tests cannot reach 0.05 unless every comparison is won; the registered rule for this case checks that the mean difference keeps its sign and that In-Context Gibbs still wins on at least 60% of the remaining datasets, which holds at both levels.

Rounding in Table 1. The two PLL rows of Study 2 against tuned CC are different results that agree to three decimals: +0.0125 [+0.0065, +0.0192] at $q = 0.3$ and +0.0121 [+0.0056, +0.0193] at $q = 0.6$.

Table 10: Study 2 without the flagged datasets.

Setting	Comparator	datasets	Δ (all datasets)	Δ (flagged removed)	W/T/L	p_{Holm}	Δ vs naive
LLP, bag 8	DLLP (linear)	16	+0.052	+0.051	15/0/1	<0.001	+0.124
PLL, $q = 0.3$	CC, tuned	7	+0.012	+0.016	6/1/0	0.062	+0.023
PLL, $q = 0.6$	CC, tuned	7	+0.012	+0.013	5/1/1	0.062	+0.168

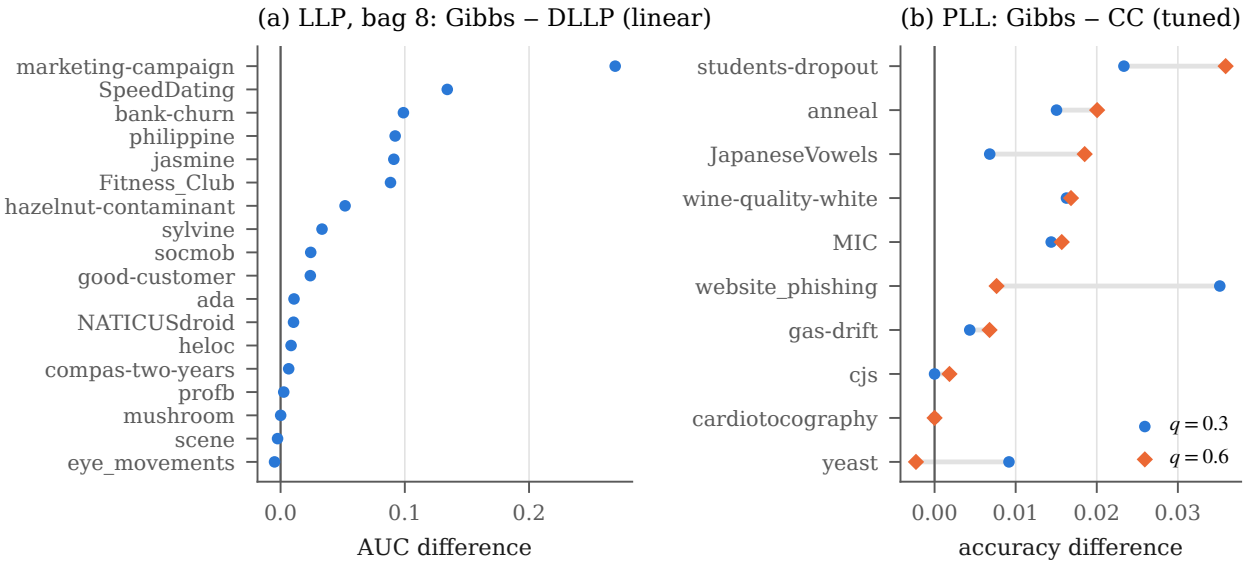


Figure 3: Study 2, per dataset: In-Context Gibbs (single chain) minus the primary comparator (positive favours In-Context Gibbs), mean over 3 seeds. (a) LLP with bags of 8, AUC against linear DLLP. (b) PLL, accuracy against tuned CC at two candidate-set rates; a bar joins the two rates of a dataset.

All baselines. Against each baseline separately (Holm over all baseline–level pairs of a setting), In-Context Gibbs is significantly better than all ten LLP baselines; on PLL every mean difference is positive and 7 of 16 pairs are significant after this stricter correction (Table 11). One baseline (default PRODEN) produced non-finite probabilities on 5 of 6 gas-drift cells, also on a rerun; its means use the remaining cells.

Table 11: Study 2, In-Context Gibbs against every baseline. Bold: significant in favour of In-Context Gibbs.

Setting	Baseline	datasets	Δ	W/T/L	p_{Holm}
LLP, bag 8 (AUC)	DLLP (MLP)	18	+0.057	17/0/1	**<0.001**
	DLLP (MLP), tuned	18	+0.041	16/0/2	**<0.001**
	DLLP (linear)	18	+0.052	16/0/2	**<0.001**
	DLLP (linear), tuned	18	+0.046	16/0/2	**<0.001**
	LLP-VAT	18	+0.062	16/0/2	**<0.001**
	LLP-VAT, tuned	18	+0.042	16/0/2	**<0.001**
	EasyLLP	18	+0.116	18/0/0	**<0.001**
	EasyLLP, tuned	18	+0.069	17/0/1	**<0.001**
	GLWS	18	+0.056	17/1/0	**<0.001**
	GLWS, tuned	18	+0.051	18/0/0	**<0.001**
PLL, $q = 0.3$ (acc.)	PRODEN	9	+0.048	9/0/0	**0.043**
	PRODEN, tuned	10	+0.027	8/1/1	0.070
	GLWS/ILL	10	+0.048	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.014	10/0/0	**0.031**
	RC	10	+0.034	9/0/1	**0.043**
	RC, tuned	10	+0.013	8/1/1	0.070
	CC	10	+0.035	10/0/0	**0.031**
	CC, tuned	10	+0.012	8/2/0	0.070
PLL, $q = 0.6$ (acc.)	PRODEN	10	+0.173	10/0/0	**0.031**
	PRODEN, tuned	10	+0.031	8/0/2	0.070
	GLWS/ILL	10	+0.128	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.015	8/0/2	0.070
	RC	10	+0.048	8/0/2	0.070
	RC, tuned	10	+0.012	8/1/1	0.070
	CC	10	+0.047	8/0/2	0.070
	CC, tuned	10	+0.012	8/1/1	0.070

Table 12: Study 2, LLP with bags of 8: AUC per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

Dataset	Gibbs	hard-EM	naive	DLLP-lin	DLLP tuned	LLP-VAT tuned	oracle
ada	**0.898**	0.895	0.742	0.888	0.880	0.883	0.912
Bank_Customer_Churn	0.849	**0.855**	0.785	0.750	0.837	0.834	0.877
compas-two-years	**0.683**	0.671	0.539	0.677	0.650	0.662	0.692
eye_movements	0.564	**0.589**	0.538	0.569	0.549	0.548	0.701
Fitness_Club	**0.831**	0.784	0.609	0.743	0.719	0.753	0.831
hazelnut-contaminant	**0.993**	0.989	0.853	0.941	0.953	0.951	0.995
heloc	**0.797**	0.788	0.759	0.789	0.788	0.787	0.813
Is-this-a-good-customer	0.633	0.634	0.576	0.610	**0.656**	0.631	0.737
jasmine	**0.870**	0.841	0.780	0.779	0.807	0.792	0.902
Marketing_Campaign	**0.878**	0.870	0.667	0.609	0.630	0.635	0.915
mushroom	**1.000**	**1.000**	0.721	1.000	0.996	1.000	1.000
NATICUSdroid	0.983	**0.985**	0.911	0.973	0.976	0.977	0.988
philippine	0.774	**0.781**	0.711	0.681	0.699	0.700	0.934
profb	0.580	0.495	0.507	0.578	0.569	**0.587**	0.675
scene	0.971	**0.983**	0.823	0.973	0.977	0.973	0.991
socmob	**0.980**	0.979	0.794	0.956	0.950	0.947	0.987
SpeedDating	**0.842**	0.752	0.768	0.708	0.783	0.745	0.871
sylvine	0.993	**0.996**	0.955	0.960	0.965	0.958	0.997
mean	0.840	0.827	0.724	0.788	0.799	0.798	0.879

Table 13: Study 2, PLL: accuracy per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

	$q = 0.3$						$q = 0.6$					
Dataset	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle
anneal	**0.991**	0.989	0.955	0.976	0.972	0.996	**0.977**	0.945	0.801	0.957	0.962	0.996
cardiotocography	**1.000**	**1.000**	0.994	**1.000**	**1.000**	1.000	**1.000**	0.886	0.476	**1.000**	**1.000**	1.000
cjs	0.997	0.997	0.962	0.997	**0.998**	1.000	**0.993**	0.830	0.391	0.991	0.989	1.000
gas-drift	0.996	**0.997**	0.993	0.991	0.992	0.998	**0.995**	0.994	0.930	0.988	0.986	0.998
JapaneseVowels	0.991	**0.995**	0.968	0.984	0.980	0.998	**0.989**	0.983	0.785	0.970	0.973	0.998
MIC	**0.887**	0.886	0.883	0.873	0.872	0.891	**0.886**	0.877	0.848	0.870	0.871	0.891
students-dropout	0.777	**0.782**	0.775	0.754	0.754	0.786	**0.770**	0.764	0.744	0.734	0.733	0.786
website_phishing	**0.823**	0.820	0.792	0.787	0.787	0.844	0.786	**0.787**	0.735	0.778	0.778	0.844
wine-quality-white	0.563	**0.571**	0.538	0.547	0.546	0.585	**0.556**	0.510	0.481	0.539	0.537	0.585
yeast	**0.597**	0.578	0.543	0.588	0.591	0.606	0.561	0.456	0.275	**0.563**	0.563	0.606
mean	0.862	0.862	0.840	0.850	0.849	0.870	0.851	0.803	0.647	0.839	0.839	0.870

Table 14: Study 2: mean score of every method (datasets averaged over 3 seeds). PRODEN (default) failed on gas-drift in 5 of 6 cells (non-finite probabilities, reproduced on a rerun); its $q = 0.3$ mean is over 9 datasets and its $q = 0.6$ gas-drift entry uses one seed.

LLP, bag 8 (AUC, 18 datasets)		
Method	default	tuned
In-Context Gibbs (ours)	0.840	
Hard-EM (same TFM)	0.827	
Stochastic EM (same TFM)	0.835	
Naive TFM	0.724	
DLLP (MLP)	0.783	0.799
DLLP (linear)	0.788	0.794
LLP-VAT	0.778	0.798
EasyLLP	0.724	0.771
GLWS	0.784	0.789
Oracle (true labels)	0.879	

PLL (accuracy, 10 datasets)				
Method	$q = 0.3$		$q = 0.6$	
	default	tuned	default	tuned
In-Context Gibbs (ours)	0.862		0.851	
Hard-EM (same TFM)	0.862		0.803	
Stochastic EM (same TFM)	0.858		0.843	
Naive TFM	0.840		0.647	
PRODEN	0.800	0.836	0.678	0.820
GLWS	0.814	0.848	0.724	0.837
RC	0.828	0.849	0.803	0.839
CC	0.827	0.850	0.804	0.839
Oracle (true labels)	0.870		0.870	

E Study 3: protocol, datasets and further results

Protocol. The protocol registers the hypotheses as H1–H3 and the criteria as C4 and C5; we write them with a double prime to keep them apart from those of Studies 1 and 2. It was committed before the code of the study, before the scoring script and before any Study 3 table was downloaded, and the scoring script was committed before any result existed. Nothing of the method was changed on these datasets: the chain settings are those of Section 4, the three starts and the equal weights were chosen on development data (Appendix G). A (TFM, dataset) pair counts if all three seeds ran without error; all 71 pairs did (213 cells). The protocol also recorded our expectations: that both hypotheses and all three criteria would hold. Two did not.

Datasets. Candidates were taken, from metadata only, from the suites of Study 2 and OpenML-CC18, after excluding 176 dataset identifiers: every dataset used in Study 1, Study 2 or development, and the 123 datasets on which TabDPT was pretrained. The size and class-balance criteria of Study 2 were applied at their most relaxed registered tier (500 to 20,000 rows, at most 2,000 features, 3 to 20 classes, a minority class of at least 5% for binary tasks), which left 43 candidates; a manual review by Study 2's rules (real data only; no dataset from the same source as one already used; one dataset per source) kept 20 and removed 8 synthetic datasets, 14 that share a source with a dataset used earlier and 1 duplicate source. 6 datasets with more than 10,000 rows were subsampled to 10,000 (stratified, fixed seed) before splitting. 5 datasets were flagged in advance for group or sequence structure. Known weaknesses, recorded in the protocol: the multiclass group has only 8 datasets, which is why the tests pool both settings and per-setting results are descriptive; three datasets had been on the reserve list of Study 2 (their tables had been hashed, no method had been run on them); and we cannot tell whether the developers of TabPFN-3 or TabICL v2 used any of these datasets.

Execution. Three sessions (10–12 November 2026). One change was made to the pre-written scoring script before any scoring: a download error at the dataset-hash step is treated as "not checked" instead of as a failed criterion (no dataset was affected). The order of the GPU queues was rebalanced between sessions. One interim look at 60 of the 71 pairs was taken on 11 November with the pre-written script; nothing was changed afterwards. The files of completed cells were byte-identical across sessions.

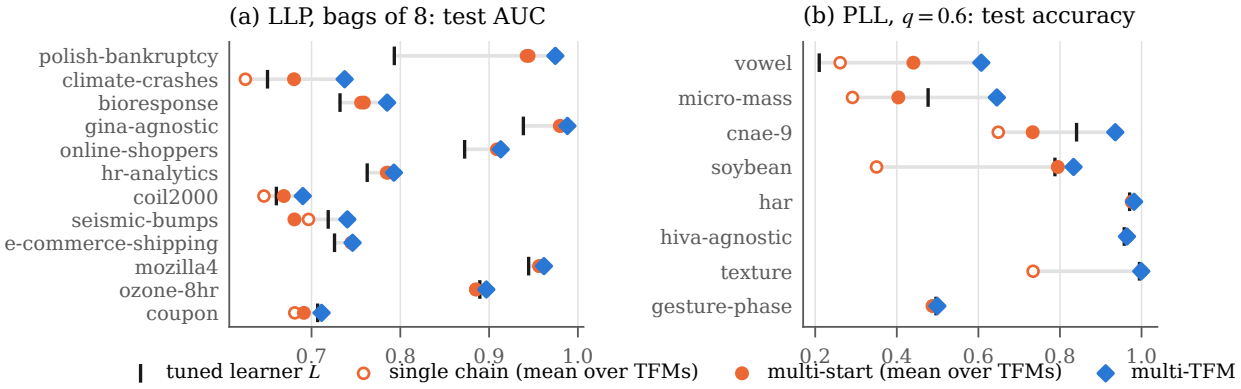


Figure 4: Study 3, per dataset (mean over 3 seeds), ordered by the gain of the multi-TFM average over the tuned learner L. "Single chain" and "multi-start" are means over the TFMs run on the dataset.

Regressions. Most pairs of Table 17 are of two kinds: single chains that stall so deeply that the average of three starts does not recover (cnae-9, micro-mass), and datasets on which L is better than every chain of that TFM (coupon, seismic-bumps, climate-crashes); the table lists every pair. The single chain itself was more than 0.02 below L in 14 of the 71 pairs. In 4 cells the single chain had an AUC below 0.5 on LLP, i.e. it converged to the labelling with the two classes exchanged within bags' counts.

Exploratory decomposition. Table 18 separates the ingredients of the two averages. Within one TFM, everything that includes the learner start or the learner's prediction is close to the registered three-start

Table 15: Study 3, per dataset: test score of the single chain ("1") and the multi-start average ("ms") for each TFM, the multi-TFM average and the tuned learner L (mean over 3 seeds). n is the number of rows after de-duplication and subsampling. TabPFN v2 ran only within its published limits.

					TabICL v2		TabPFN-3		TabDPT		TabPFN v2			
ID	Dataset	n	d	C	1	ms	1	ms	1	ms	1	ms	multi-TFM	L
LLP, bags of 8 (AUC)														
1038	gina-agnostic	3,468	970	2	0.993	0.993	0.990	0.990	0.956	0.959	–	–	0.988	0.939
1046	mozilla4	10,000	5	2	0.956	0.955	0.961	0.962	0.957	0.959	0.951	0.950	0.962	0.945
1467	climate-crashes	540	20	2	0.690	0.731	0.708	0.714	0.417	0.528	0.687	0.748	0.737	0.650
1487	ozone-8hr	2,526	72	2	0.870	0.881	0.901	0.892	0.895	0.899	0.875	0.868	0.897	0.890
4134	bioresponse	3,751	1776	2	0.775	0.774	0.724	0.723	0.768	0.780	–	–	0.785	0.732
46916	coil2000	8,380	85	2	0.654	0.679	0.653	0.662	0.645	0.650	0.634	0.683	0.690	0.660
46924	e-commerce-shipping	10,000	10	2	0.748	0.746	0.746	0.746	0.742	0.744	0.746	0.743	0.746	0.726
46935	hr-analytics	10,000	12	2	0.792	0.794	0.788	0.793	0.783	0.782	0.775	0.774	0.793	0.763
46937	coupon	10,000	24	2	0.708	0.711	0.692	0.697	0.666	0.682	0.659	0.676	0.711	0.707
46947	online-shoppers	10,000	17	2	0.906	0.907	0.909	0.919	0.907	0.907	0.912	0.912	0.913	0.872
46950	polish-bankruptcy	5,850	64	2	0.973	0.975	0.978	0.978	0.948	0.951	0.870	0.874	0.975	0.793
46956	seismic-bumps	2,578	15	2	0.554	0.491	0.755	0.731	0.746	0.749	0.731	0.751	0.740	0.719
PLL, $q = 0.6$ (accuracy)														
42	soybean	631	35	19	0.589	0.798	0.074	0.796	0.388	0.789	–	–	0.833	0.788
307	vowel	990	12	11	0.476	0.586	0.091	0.396	0.217	0.340	–	–	0.607	0.210
1468	cnae-9	1,052	856	9	0.920	0.930	0.141	0.377	0.885	0.892	–	–	0.936	0.841
1478	har	10,000	561	6	0.985	0.984	0.975	0.975	0.969	0.969	–	–	0.982	0.971
1515	micro-mass	571	1300	20	0.548	0.671	0.047	0.221	0.279	0.320	–	–	0.645	0.477
4538	gesture-phase	9,873	32	5	0.507	0.507	0.470	0.470	0.483	0.484	0.488	0.494	0.499	0.496
40499	texture	5,473	40	11	1.000	1.000	0.205	0.999	0.998	0.999	–	–	0.999	0.995
46933	hiva-agnostic	3,825	1617	3	0.964	0.964	0.964	0.964	0.964	0.964	–	–	0.964	0.958

Table 16: Study 3 by TFM. G: mean of multi-start − single chain; regr.: pairs counted by C4″; AUC(H3″): the diagnostic within the TFM. Last three columns (exploratory): mean difference over datasets and W/T/L.

TFM	datasets	G	regr.	AUC(H3)	multi-TFM − multi-start	multi-start − L	single − L
TabICL v2	20	+0.024	1	0.85	+0.016 (10/5/5)	+0.047 (18/0/2)	+0.024 (16/0/4)
TabPFN-3	20	+0.112	4	0.93	+0.070 (12/5/3)	−0.006 (15/0/5)	−0.118 (11/0/9)
TabDPT	20	+0.037	3	0.59	+0.053 (16/2/2)	+0.011 (14/0/6)	−0.026 (13/0/7)
TabPFN v2	11	+0.013	2	0.64	+0.017 (8/1/2)	+0.023 (8/0/3)	+0.010 (7/0/4)

average, and everything that does not is close to the single chain; adding L's prediction to the three-start average gives +0.021 ($p = 0.005$). Across TFMs, the average of the chains started from L equals the registered multi-TFM average (−0.001), the average of single chains is −0.028 below it, and TabPFN v2, available on 11 datasets, makes no difference. With TabICL v2 alone, the configuration of Study 2, the single chain was above L by +0.019 on LLP (9/0/3) and +0.032 on PLL (7/0/1), and its multi-start average by +0.020 and +0.088 (10/0/2 and 8/0/0); here too single chains stalled on individual datasets (Table 15). These comparisons were chosen after seeing the confirmatory results and carry no error control.

Which configuration, on what evidence. For a reader deciding what to run, Table 19 separates the three configurations by the kind of evidence behind them.

Table 17: The 10 (TFM, dataset) pairs in which the multi-start average is more than 0.02 below the better of the single chain and L (criterion C4″), ordered by the size of the regression.

TFM	Dataset	single	multi-start	L
TabPFN-3	cnae-9	0.141	0.377	0.841
TabPFN-3	micro-mass	0.047	0.221	0.477
TabICL v2	seismic-bumps	0.554	0.491	0.719
TabDPT	micro-mass	0.279	0.320	0.477
TabDPT	climate-crashes	0.417	0.528	0.650
TabPFN v2	coupon	0.659	0.676	0.707
TabPFN-3	gesture-phase	0.470	0.470	0.496
TabDPT	coupon	0.666	0.682	0.707
TabPFN-3	seismic-bumps	0.755	0.731	0.719
TabPFN v2	ozone-8hr	0.875	0.868	0.890

Table 18: Exploratory decomposition of Study 3 (not registered; computed after the confirmatory scoring from the stored predictions, no chain was rerun). Mean test score over datasets (within-TFM rows: mean over the TFMs of each dataset), and difference to the registered average with W/T/L over the 20 datasets and a two-sided Wilcoxon signed-rank p, unadjusted (SciPy defaults: zero differences dropped; exact distribution without zeros, normal approximation otherwise).

Prediction (one TFM)	LLP	PLL	− multi-start	W/T/L	p
single chain (warm start)	0.801	0.589	−0.058	1/6/13	0.002
chain started from L	0.799	0.740	+0.001	5/6/9	0.398
chain with cold start	0.793	0.589	−0.063	0/4/16	<0.001
multi-start (registered: warm, L, cold)	0.807	0.725			
warm and L starts	0.805	0.731	+0.001	5/13/2	0.334
warm and cold starts (no L)	0.806	0.601	−0.050	1/6/13	<0.001
single chain averaged with L's prediction	0.813	0.737	+0.008	12/1/7	0.261
multi-start averaged with L's prediction	0.816	0.765	+0.021	14/2/4	0.005
tuned learner L	0.783	0.717	−0.018	6/0/14	0.143
Prediction (all TFMs)	LLP	PLL	− multi-TFM	W/T/L	p
multi-TFM (registered)	0.828	0.808			
single chains only, averaged over TFMs	0.823	0.747	−0.028	3/8/9	0.020
warm and cold chains only (no L)	0.829	0.758	−0.020	2/10/8	0.044
chains started from L only	0.823	0.814	−0.001	5/7/8	0.658
without TabPFN v2	0.827	0.808	−0.001	3/13/4	0.286

Table 19: The three configurations, the evidence for each and their cost. "Confirmatory": a pre-registered test passed; "exploratory": computed after the registered scoring, without error control.

Configuration	Against specialised learners	Against simpler variants	Cost
Single chain, one TFM	confirmatory with TabICL v2 (Study 2); criteria missed with TabPFN v2 (Study 1)	confirmatory: lower log loss than hard-EM	1 chain
Multi-start, one TFM	registered descriptive test not significant (+0.018, $p = 0.14$); 9 of 71 pairs more than 0.02 behind L	confirmatory: above the single chain (H1″); exploratory: no better than the chain started from L	3 chains and tuning L
Multi-TFM	exploratory: above L on 20/0/0 datasets (W/T/L)	confirmatory: above the average TFM (H2″), level with the best (C5″)	9 or 12 chains and tuning L

F Supplementary analysis S3B: protocol and further results

Protocol and timeline. The plan and its scoring script were committed on 12 November 2026, after Study 3 had been scored and before any of its runs; the code that ran was committed six minutes later, and the hashes of all 87 code files in every run match it. Three parts: B1, the two non-neural baselines on CPU, with two tests (multi-TFM average against each, Holm-adjusted) and one comparison of L with the better of them; B2, the naive TFM, hard-EM and the oracle with every TFM of Study 3, with one test (single chain against hard-EM on log loss); B3, the warm-start chain for 100 sweeps on the PLL cells of TabICL v2, TabPFN-3 and TabDPT, with prediction P22. Every row was checked against the split and weak-label hashes of Study 3; none differed. *Deviation from the plan's schedule:* the GPU part needed two Kaggle sessions. After the first, 5 of the 72 cells of B3 were missing; B1 and B2 were scored, B3 was scored on the one TFM with all its cells as the plan prescribes for incomplete runs, and the remaining cells were looked at once (recorded in the experiment log). The second session added exactly the 5 missing cells, leaving every earlier row unchanged, and B3 was scored again on all 72 cells. The interim look had already shown that the second part of P22 would fail whatever the missing cells gave.

Baselines. GBDT self-training uses histogram gradient boosting with default hyperparameters: for PLL, pseudo-labels restricted to each point's candidate set, reassigned out-of-fold for 3 rounds; for LLP, an out-of-fold regression of the bag proportion as the initial score, then 3 rounds in which the top-c_g points of each bag (by current score) are labelled positive. PL-kNN predicts by a vote over the candidate sets of the $k = 10$ nearest training points. Neither was tuned.

Table 20: B2: mean test score over datasets by TFM (AUC for LLP, accuracy for PLL; number of datasets in brackets).

TFM	Setting (n)	naive	hard-EM	single chain	multi-start	oracle
TabICL v2	LLP (12)	0.707	0.794	0.802	0.803	0.876
TabICL v2	PLL (8)	0.442	0.678	0.749	0.805	0.941
TabPFN-3	LLP (12)	0.729	0.796	0.817	0.817	0.880
TabPFN-3	PLL (8)	0.357	0.367	0.371	0.650	0.940
TabDPT	LLP (12)	0.716	0.814	0.786	0.799	0.874
TabDPT	PLL (8)	0.382	0.577	0.648	0.720	0.925
TabPFN v2	LLP (10)	0.726	0.781	0.784	0.798	0.858
TabPFN v2	PLL (1)	0.447	0.526	0.488	0.494	0.745

Table 21: B3: chains run for 100 instead of 30 sweeps on PLL ($q = 0.6$). Stuck: the 30-sweep chain was more than 0.02 below the better of the multi-start average and L (Study 3). Repaired: the 100-sweep chain is within 0.02 of it. Gain: mean test accuracy after 100 minus after 30 sweeps. Label accuracy: mean over cells.

TFM	cells	stuck	repaired	gain (all)	label acc. sweep 30 → 100
TabICL v2	24	9	3	+0.031	0.76 → 0.78
TabPFN-3	24	16	1	+0.111	0.40 → 0.49
TabDPT	24	10	1	+0.016	0.63 → 0.65
All	72	35	5	+0.053	

G Development between Study 2 and Study 3

The multi-start and multi-TFM samplers were developed between 4 and 10 November 2026 on the datasets of Studies 1 and 2 and on development datasets. All of it is exploratory. To keep ourselves honest we wrote the plan of every run, the analysis script and numbered predictions with thresholds before the run; Table 23 lists every prediction and its outcome, including the 7 of 17 that were wrong (the number P10 was never assigned).

Table 22: B1 per dataset (mean over 3 seeds; TFM-based columns averaged over the TFMs of each dataset).

Setting	Dataset	GBDT	PL-kNN	single	multi-start	L	multi-TFM
LLP	1038	0.949	–	0.979	0.980	0.939	0.988
LLP	1046	0.959	–	0.956	0.957	0.945	0.962
LLP	1467	0.654	–	0.625	0.680	0.650	0.737
LLP	1487	0.893	–	0.885	0.885	0.890	0.897
LLP	4134	0.763	–	0.756	0.759	0.732	0.785
LLP	46916	0.638	–	0.646	0.669	0.660	0.690
LLP	46924	0.743	–	0.745	0.745	0.726	0.746
LLP	46935	0.781	–	0.785	0.786	0.763	0.793
LLP	46937	0.708	–	0.681	0.691	0.707	0.711
LLP	46947	0.908	–	0.909	0.911	0.872	0.913
LLP	46950	0.934	–	0.942	0.944	0.793	0.975
LLP	46956	0.621	–	0.697	0.681	0.719	0.740
PLL	42	0.130	0.679	0.350	0.795	0.788	0.833
PLL	307	0.270	0.365	0.261	0.441	0.210	0.607
PLL	1468	0.373	0.522	0.649	0.733	0.841	0.936
PLL	1478	0.966	0.878	0.976	0.976	0.971	0.982
PLL	1515	0.122	0.209	0.291	0.404	0.477	0.645
PLL	4538	0.534	0.448	0.487	0.489	0.496	0.499
PLL	40499	0.799	0.915	0.734	0.999	0.995	0.999
PLL	46933	0.959	0.945	0.964	0.964	0.958	0.964

Table 23: Predictions written before each development run, and outcomes. "Regression": more than 0.02 below the better of the single chain and the tuned learner L. P1–P4 belong to a first phase in which one chain was selected by weak-label fit; P4 failed and the selection rule was replaced by averaging.

	Prediction (written before the run)	Result	Outcome
P1	a chain of TabPFN v2 started from tuned CC ends above 0.80 label accuracy in 9/9 runs and is within 0.02 of CC	0.95–0.99; above CC on 3/3 datasets	correct
P2	on semeion, where the gain is negative at 0.85–0.95, that chain still ends below 0.50	ends at 0.76–0.81	wrong
P3	LLP, TabPFN v2: where the gain is negative at the start, the chain ends below its start (2 datasets)	2/2 datasets	correct
P4	choosing the chain by weak-label fit picks a near-best start in $\geq 80\%$ of cells	19/24 (79.2%)	wrong
P5	stacking with L has fewer regressions than equal averaging (TabPFN v2)	1 regression each	wrong
P6	aggregators differ by < 0.005 in mean gain (TabICL v2)	max. gap 0.0031	correct
P7	the chosen average is not below the single chain by more than 0.002 in any of four groups	$+0.0019$ to $+0.0043$	correct
P8	chain-seeded map predicts final label accuracy: median error ≤ 0.10	0.016 (234 cells)	correct
P9	chain-seeded map is more accurate than the uniform map	0.016 vs. 0.073	correct
P11	TabPFN-3 has no low attractor on the four stalling datasets	0/4 datasets	wrong
P12	multi-start at equal cost (3×10 sweeps) still gains, ≤ 3 regressions	6 regressions	wrong
P13	four-TFM average within 0.01 of the best TFM on average, > 0.02 behind on ≤ 2 datasets (seed 0)	-0.008; 2 worse by > 0.02	correct
P14	TabPFN-3 multi-start beats TabPFN v2 multi-start by 0.05 on $\geq 3/4$ stalling datasets	1/4	wrong
P15	disagreement of three 3-sweep probe chains predicts basin splits (AUC ≥ 0.75)	AUC 0.45	wrong
P16	same on 23 datasets, 3 seeds (≤ 3 behind), and not below any single TFM on average	-0.004; 1 worse by > 0.02	correct
P17	multi-start gains with TabPFN-3 and TabDPT, ≤ 2 regressions each	$+0.0115$ / $+0.0002$; 2 / 2	correct
P18	post-chain diagnostic: AUC ≥ 0.80 on new development cells	AUC 0.86 (138 cells)	correct

Choice of the average. Three ways of combining the three chains were compared on 78 (TFM, dataset) pairs (234 cells; TabICL v2 and TabPFN v2; four LLP datasets of TabPFN v2 are missing because of download errors): equal weights, and two stacking rules that use weak-label fit. The registered rule chose the combination with the fewest regressions: equal weights (2 against 3 and 3). On these pairs the equal-weight average gained $+0.0032$ over the single chain on average (better on 39, worse on 11, tied on 28) and was never more than 0.02 below it. This small gain is why we expected a typical dataset of Study 3 to gain little, which held (median $+0.003$), and few pairs to regress, which did not. Running the three chains for a third of the sweeps each, at the cost of one chain, lost the gain (P12).

TFMs. TabPFN-3 was preferred to its successor by a registered rule on speed (the latter was 3.1 times slower per call on a T4). Under uniformly placed errors TabPFN-3 keeps the low attractor of TabPFN v2 on the four stalling datasets of Study 1 (P11), yet its chains started from L reach high accuracy on them, and so do those of TabPFN v2 on three of the four (P14 was wrong for that reason): stalling is a matter of the start as much as of the TFM. On 23 development and earlier datasets the four-TFM average was -0.004 from the best TFM of each dataset (P16), which fixed criterion C5″.

Response maps. For every cell of the earlier runs we measured the one-sweep response at a grid of accuracy levels, placing errors either uniformly or according to the stored label marginals of the chain (wrong points chosen with probability proportional to one minus the marginal of the true label), and predicted the final

label accuracy by a rule fixed in advance: start at the chain's initial accuracy and follow the sign of the interpolated gain to the next crossing. Table 24 and Figure 5 give the errors. The 90th percentile of the error with chain-seeded maps is 0.051.

Table 24: Median absolute error of the predicted final label accuracy, by the way errors are placed when the response map is measured ("structured": according to the warm start, PLL only).

TFM	Setting	cells	chain-seeded	uniform	structured
TabICL v2	LLP	96	0.026	0.081	–
TabICL v2	PLL	69	0.011	0.042	0.053
TabPFN v2	LLP	30	0.025	0.070	–
TabPFN v2	PLL	39	0.009	0.083	0.079
All		234	0.016	0.073	

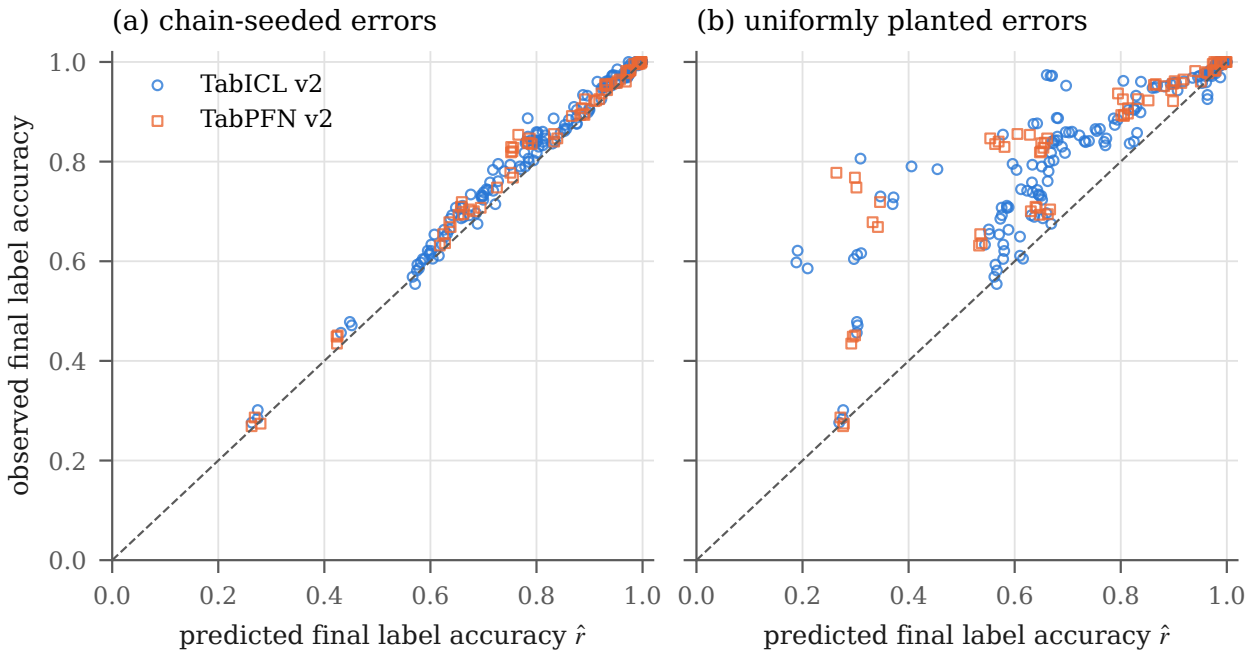


Figure 5: Final label accuracy of the warm-start chains of 234 development cells against the fixed point predicted from a response map, with errors placed (a) by the chain's own label marginals, (b) uniformly.

Diagnostics. A first label-free signal, the disagreement of three short probe chains, carried no information (P15). The second, the disagreement of the warm-start and learner-start chains, was proposed on old runs, held on new development cells (P18, AUC 0.86 on 138 cells) and then failed its registered threshold in Study 3 (H3″).

Harder settings, not pursued. With bags of 32 every TFM had cells in which the chain converged to exchanged classes, and the multi-start average repaired them only in part. Candidate sets whose distractors depend on the instance (drawn in proportion to a classifier's probabilities), run with TabICL v2 only, were much harder than uniform ones. Bags formed by clustering the features were easier than random bags and were dropped. These settings are left to future work.

H Positioning against the closest works

Table 25: Positioning against the closest works. "Partial": exact posterior over labels inside an EM loop, with conditionally independent predictions. *The TFM is frozen; the LLP warm start trains a linear DLLP model and the learner start a tuned specialised learner (Section 4).

	no training (frozen model)	tabular data	several weak-label types	inference over latent labels with guarantees	dependence among latent labels
In-Context PU (Liu et al., 2026)	– (pretrains a PFN)	✓	– (PU only)	–	–
GLWS (Chen et al., 2024b), ILL (Chen et al., 2024a)	–	–	✓	partial	–
TabPFN imputation (Sepin, 2026)	✓	✓	– (missing outcome)	–	–
Noise immunity (Hu & Ghelichi, 2026)	✓	✓	– (label noise)	– (measurement)	–
In-Context Gibbs (ours)	✓*	✓	✓	✓	✓

I Correctness experiments

We enumerate every labelling of small problems ($n \leq 14$), build the exact transition matrix of each kernel with an analytic PPD as q, and compare its stationary distribution with π. Two priors are used: one in which features carry no information about the label mechanism, and a three-component mixture in which they do. Across all 14 configurations the TV distance is at most 3×10^{-14} (Table 26), and running the actual sampler for 10^5 sweeps reaches the Monte Carlo floor of i.i.d. sampling. For Theorem 2 we use conditionals with deliberate, incompatible perturbations (30 configurations) and small PFNs trained on the same prior; the bound held in 42 of 42 configurations and was 1.9 to 15 times loose (median 3.9; Table 27); a fully trained PFN gives TV 0.012 (single-site) and 0.007 (swap). On an exactly solvable pooled problem with $n = 1{,}000$ and $K = 10$, blocking over all points changes predictions by 1.7×10^{-4}, whereas blocking only the free points biases them by 0.031, which is why blocks include fixed points.

Table 26: Exactness (Theorem 1, Proposition 2(a)): TV between the stationary distribution of the exact kernel and π.

Prior	Weak labels	Kernel	n	states	TV(stationary, π)
cell	PU-SCAR (c=0.5)	single	10	128	3.3e-15
cell	SSL	single	10	128	4.0e-15
cell	PLL	single	10	128	4.0e-15
cell	Noisy (T known)	single	8	256	1.9e-14
cell	LLP (3 bags of 4)	swap	12	96	3.1e-15
cell	SSL, block-MH (m=3)	block-mh	10	128	2.8e-14
mixture	PU-SCAR (c=0.5)	single	10	128	3.3e-15
mixture	SSL	single	10	128	3.0e-15
mixture	PLL	single	10	128	3.0e-15
mixture	Noisy (T known)	single	8	256	2.3e-14
mixture	LLP (3 bags of 4)	swap	12	96	4.8e-15
mixture	SSL, block-MH (m=3)	block-mh	10	128	4.0e-15
cell	PLL (C=3, |S_i|=2)	single	7	128	1.2e-14
mixture	PLL (C=3, |S_i|=2)	single	7	128	3.8e-15

Table 27: Stability (Theorem 2): actual TV between the stationary distributions of the approximate and exact chains, the bound $\tau(P)\bar{\varepsilon}$, and their ratio (looseness).

Conditionals	Kernel	configs	median TV($\tilde{\pi}, \pi$)	median bound	bound held	median looseness
perturbed PPD (incompatible)	single	15	0.0181	0.246	15/15	9.8
perturbed PPD (incompatible)	swap	15	0.0254	0.066	15/15	2.2
small PFN (trained)	single	6	0.1419	0.628	6/6	4.9
small PFN (trained)	swap	6	0.0480	0.107	6/6	3.3

J Ablations

Table 28: Mixing and the exact sampler, by TFM. TabPFN: Study 1 (registered ablations); TabICL: exploratory lot on the Study 1 datasets. "3 small": the three smallest datasets of each group, as registered for $\widehat{R}$. Columns 2–5: medians over cells of the fraction of training labels with $\widehat{R} > 1.1$ (two chains) and of the absolute difference between the test metrics of the two chains. Columns 6–7: mean difference in AUC (LLP) or accuracy (PLL) between the MH-corrected and the default sampler at 30 sweeps, with the mean MH acceptance rate. In the last row, $\widehat{R}$ uses all four datasets with ≥8 classes, and the MH comparison only two of them (mfeat-factors, optdigits; 5 of 12 cells), because the corrected sampler ran out of the 12-hour compute session on optdigits.

	labels with $\widehat{R} > 1.1$		test-metric gap, 2 chains		MH-corrected − default	
Setting (datasets)	TabPFN	TabICL	TabPFN	TabICL	TabPFN	TabICL [acc. rate]
LLP, bag 8 (3 small)	11%	28%	0.041	0.018	+0.010	+0.033 [0.68]
PLL, $q = 0.3$ (3 small)	6%	6%	0.006	0.018	+0.006	−0.009 [0.78]
PLL, $q = 0.6$ (3 small)	21%	31%	0.016	0.021	–	−0.021 [0.72]
PLL, $q = 0.6$, ≥8 classes (4; MH: 2)	–	22%	–	0.002	–	−0.016 [0.47]

Table 29: Study 2 confirmatory tests with a second chain per cell (exploratory lot run after the analysis; same data, weak labels and start, other sampling seed). Each entry: mean difference, W/T/L over datasets, p_{Holm} (Holm within each hypothesis, as registered). Chain a is the registered result. The averaged column uses the stored test predictions of both chains (float16).

Test	chain a (registered)	chain b	chains a and b averaged
H2′: LLP, bag 8, vs DLLP (linear)	+0.052, 16/0/2, <0.001	+0.046, 15/0/3, 0.002	+0.052, 17/0/1, <0.001
H4′: PLL, $q = 0.3$, vs CC (tuned)	+0.012, 8/2/0, 0.016	+0.013, 9/1/0, 0.008	+0.013, 9/1/0, 0.008
H4′: PLL, $q = 0.6$, vs CC (tuned)	+0.012, 8/1/1, 0.016	+0.011, 8/1/1, 0.074	+0.013, 8/1/1, 0.008

Table 30: In-Context Gibbs versus hard-EM with the same TFM and warm start, log loss. A cell is a (dataset, weak-label level) pair; Bayesian signed-rank test with ROPE ±0.005; Spearman correlation between weakness level and the gap. †Computed on PLL cells (LLP has a single level), as stated before the analysis was run.

Study (TFM; settings)	cells	Gibbs better	P(Gibbs better)	P(ROPE)	Spearman ρ
Study 1 (TabPFN v2; all six settings)	128	86.7%	1.00	0.00	0.36
Study 2 (TabICL v2; LLP bag 8, PLL)	38	78.9%	1.00	0.00	0.66†

Table 31: Study 1 ablations (TabPFN v2): each variant against the default on the same cells. Blocks, sweeps and MH: the three smallest datasets per group; transductive and cold start: all datasets.

Variant (vs. default)	Setting	cells	Δ AUC/acc.	W/T/L	Δ log loss (+ = better)
$K = 20$	LLP, bag 8	9	+0.007	5/0/4	+0.024
$K = 20$	PLL, 0.3	9	+0.006	5/2/2	−0.000
$K = 5$ (default 10)	LLP, bag 8	9	−0.005	6/0/3	+0.006
$K = 5$ (default 10)	PLL, 0.3	9	+0.004	5/1/3	+0.009
15 sweeps (default 30)	LLP, bag 8	9	+0.005	4/0/5	+0.001
15 sweeps (default 30)	PLL, 0.3	9	+0.004	4/2/3	−0.001
60 sweeps	LLP, bag 8	9	+0.014	4/0/5	−0.012
60 sweeps	PLL, 0.3	9	+0.005	4/5/0	+0.009
MH-corrected blocks (exact)	LLP, bag 8	9	+0.010	4/0/5	−0.019
MH-corrected blocks (exact)	PLL, 0.3	9	+0.006	4/3/2	+0.009
transductive prediction	LLP, bag 8	30	−0.013	7/0/23	−0.042
cold start	LLP, bag 8	30	−0.005	10/0/20	−0.002
cold start	PLL, 0.3	42	+0.000	11/23/8	+0.000
cold start	PLL, 0.6	42	+0.012	14/9/19	+0.023
PU-SAR prior $N(0, 1)$	PU-SAR, 0.1	30	−0.002	15/3/12	−0.015
PU-SAR prior $N(0, 1)$	PU-SAR, 0.3	30	+0.001	10/1/19	+0.002
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.1	30	−0.004	7/9/14	+0.000
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.3	30	−0.001	10/12/8	−0.016

K Real partial labels

Table 32: Real partial labels (Study 1, TabICL v2 for all TFM-based methods): accuracy averaged over 10 random 90/10 splits, and mean rank over all splits of the three datasets among the ten ranked methods (In-Context Gibbs, naive TFM and eight baselines). Bold: best non-oracle accuracy.

Method	BirdSong	Lost	MSRCv2	mean rank
In-Context Gibbs (ours)	**0.759**	**0.750**	**0.628**	2.15
Hard-EM (same TFM)	0.730	0.665	0.627	–
Stochastic EM (same TFM)	0.744	0.716	0.622	–
Naive TFM	0.689	0.481	0.491	7.27
PRODEN (default)	0.667	0.373	0.344	9.88
PRODEN (tuned)	0.707	0.748	0.465	5.12
GLWS (default)	0.722	0.712	0.582	3.98
GLWS (tuned)	0.731	0.747	0.490	4.03
RC (default)	0.719	0.720	0.424	6.00
RC (tuned)	0.702	0.721	0.478	5.23
CC (default)	0.719	0.717	0.425	6.02
CC (tuned)	0.700	0.723	0.479	5.32
Oracle (true labels)	0.835	0.889	0.824	–

L Datasets, baselines and tuning

Datasets. Study 1 uses every OpenML-CC18 dataset with real (non-synthetic) data, 500 to 10,000 rows, at most 500 features and, for binary tasks, a minority class of at least 10% (multiclass: 3–10 classes), one per source; Study 2 applies the same criteria to five curated suites after excluding OpenML-CC18 and Study 1 (Table 34); the selection for Study 3 is described in Appendix E. Duplicate rows are removed before splitting. TFMs receive ordinally encoded features; MLP baselines receive median-imputed, standardised features with one-hot encoded categories, fitted on the training split.

Baselines and tuning. Each baseline runs in the configuration of its original code or paper and tuned by 20 random configurations (the default among them) from the spaces below, scored by the method's own loss on a 20% validation split with weak labels only (split by bag for LLP); the best configuration is refit on the full training split. MLP-based learners (DLLP, LLP-VAT, EasyLLP, GLWS, PRODEN) search learning rate $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$, weight decay $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$, hidden layers in $\{(), (64), (128, 128)\}$ and epochs in $\{100, 200, 400\}$, plus bags per batch for LLP, VAT parameters for LLP-VAT, the optimiser for PRODEN and the entropy term for GLWS/ILL; RC and CC use a linear model with the PRODEN space (Table 33).

Table 33: Search spaces (20 random configurations, the default included). LogU: log-uniform.

Baseline	Default	Search space
DLLP (MLP, linear), EasyLLP	lr 10^{-3}, wd 0, 200 epochs	lr $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$; wd $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$; hidden $\in \{(), (64), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; bags/batch $\in \{8, 32\}$
LLP-VAT	as DLLP + VAT defaults of the official code	as DLLP; $\xi \in \{10^{-6}, 10^{-3}\}$; $\epsilon \in \{0.1, 0.5, 1, 2\}$; consistency weight $\sim \mathrm{LogU}(10^{-2}, 1)$
GLWS / ILL	official code, MLP backbone	lr, wd as DLLP; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; entropy term on/off (PLL)
PRODEN	lr 10^{-3}, wd 10^{-5}, hidden (128, 128), 200 epochs, SGD	lr, wd, hidden, epochs as DLLP; optimiser $\in$ {SGD (momentum 0.9), Adam}
RC, CC	linear model as in the original paper	as PRODEN
SAR-EM (PU-SAR)	$C_f = C_e = 1$	$C_f, C_e \sim \mathrm{LogU}(10^{-3}, 10^{3})$
nnPU, uPU, forward correction	lr 10^{-3}, wd 0, hidden (100), 200 epochs	lr $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$; wd $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$
VPU	official code	lr $\sim \mathrm{LogU}(10^{-5}, 10^{-3})$; mixup $\lambda \in \{0.01, 0.03, 0.1, 0.3\}$; $\alpha \in \{0.1, 0.3, 1\}$
Self-training	logistic regression, $C = 1$, threshold 0.75	$C \sim \mathrm{LogU}(10^{-3}, 10^{3})$; threshold $\in \{0.6, 0.75, 0.9, 0.95\}$
Label spreading	kNN, $k = 7$, $\alpha = 0.2$	kernel $\in$ {kNN, RBF}; $k \in \{3, 5, 7, 11, 15, 21\}$; $\gamma \sim \mathrm{LogU}(10^{-3}, 10)$; $\alpha \in \{0.1, 0.2, 0.5, 0.8\}$
VIME	official code	$p_m \in \{0.1, 0.3, 0.5\}$; $\alpha \in \{0.5, 1, 2\}$; $\beta \in \{0.1, 1, 10\}$; hidden $\in \{(64), (128)\}$
Confident learning + TFM	default (not tuned)	–

Table 34: Datasets (n after removing duplicates; *subsampled to 10,000 rows by the registered rule; S1: flagged for the sensitivity analysis).

Study 1 (OpenML-CC18)						Study 2 (new curated suites)					
ID	name	n	d	C	S1	ID	name	n	d	C	S1
3	kr-vs-kp	3,196	36	2		24	mushroom	8,124	22	2	
29	credit-approval	690	15	2		312	scene	2,400	299	2	
44	spambase	4,210	57	2		470	profb	672	9	2	
50	tic-tac-toe	958	9	2		934	socmob	1,156	5	2	
1049	pc4	1,344	37	2		40536	SpeedDating	8,378	120	2	✓
1489	phoneme	5,395	5	2		41143	jasmine	2,984	144	2	
1494	qsar-biodeg	1,052	41	2		41145	philippine	5,832	308	2	
6332	cylinder-bands	540	37	2		41146	sylvine	5,124	20	2	
23381	dresses-sales	500	12	2		41156	ada	4,146	48	2	
38	sick	3,711	29	2		44130	eye_movements	7,608	20	2	✓
11	balance-scale	625	4	3	✓	45026	heloc	9,444	22	2	
12	mfeat-factors	1,994	216	10		45039	compas-two-years	2,712	11	2	
28	optdigits	5,620	64	10		46911	Bank_Customer_Churn	10,000	10	2	
46	splice	3,006	60	3		46927	Fitness_Club	1,499	6	2	
182	satimage	6,430	36	6		46930	hazelnut-spread-contamin	2,400	30	2	
188	eucalyptus	736	19	5		46938	Is-this-a-good-customer	1,723	13	2	
458	analcatdata_authorship	841	70	4		46940	Marketing_Campaign	2,058	25	2	
469	analcatdata_dmft	556	4	6		46969	NATICUSdroid	7,491	86	2	
1475	first-order-theorem-prov	5,636	51	6		181	yeast	1,453	8	10	
1497	wall-robot-navigation	5,456	24	4	✓	375	JapaneseVowels	9,961	14	9	✓
1501	semeion	1,593	256	10		1466	cardiotocography	2,115	35	10	
40966	MiceProtein	1,080	77	8	✓	23380	cjs	1,811	33	6	✓
40975	car	1,728	6	4	✓	40498	wine-quality-white	3,961	11	7	
40982	steel-plates-fault	1,941	27	7		46906	anneal	886	38	5	
						46960	students_dropout_and_	4,424	36	3	
						46963	website_phishing	724	9	3	
						46980	MIC	1,699	111	8	
						1476	gas-drift	10,000*	128	6	✓

M Secondary settings of Study 1

Study 1 also registered secondary settings (Table 35). In-Context Gibbs improved strongly on the naive TFM for larger bags, as for bags of 8, but not for PU with labels selected completely at random, class-dependent noise or SSL, where its difference to the naive TFM was at most -0.013 AUC. This is the pattern Proposition 5 predicts, and it was registered in advance as exploratory analyses D1–D3: the advantage over the naive TFM grows from $q = 0.3$ to $q = 0.6$ (D1, one-sided $p = 0.039$) and from bags of 8 to 16 (D2, $p = 0.010$), and stays below $+0.02$ AUC in the settings where the naive labels lose no information (D3, all six levels). Against the primary comparators of these settings no difference was significant except class-dependent noise at $r = 0.4$, where confident learning combined with the TFM was better.

Table 35: Study 1 secondary settings (TabPFN v2). Δ positive favours In-Context Gibbs. †Log loss after matching the known class prior (sign flipped so that positive is better).

Setting	Metric	Primary comparator	In-Context Gibbs − comparator			In-Context Gibbs − naive TFM	
			Δ	W/T/L	p_{Holm}	Δ	W/T/L
LLP, bag 16	AUC	DLLP (linear)	+0.040	7/0/3	0.232	+0.206	9/0/1
LLP, bag 32	AUC	DLLP (linear)	+0.048	6/0/4	0.375	+0.223	10/0/0
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.1$	AUC	nnPU	+0.034	6/0/4	0.967	+0.007	6/0/4
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.3$	AUC	nnPU	+0.032	8/0/2	0.109	+0.017	6/0/4
PU-SCAR, $c = 0.1$	log loss†	VPU, tuned	+0.118	8/0/2	0.111	−0.033	5/0/5
PU-SCAR, $c = 0.3$	log loss†	VPU, tuned	+0.179	8/0/2	0.111	−0.028	2/0/8
Class-dependent noise, $r = 0.2$	AUC	CL + TFM	−0.006	3/0/7	0.105	−0.013	1/0/9
Class-dependent noise, $r = 0.4$	AUC	CL + TFM	−0.048	2/0/8	0.020	−0.054	1/0/9
SSL, 5% labelled	AUC	Self-training, tuned	−0.002	5/0/5	1.000	−0.041	0/0/10
SSL, 20% labelled	AUC	Self-training, tuned	−0.009	8/0/2	0.750	−0.016	0/0/10

N Statements

Data and code availability. All datasets are public: OpenML (dataset identifiers in Appendices E and L) and the PALM collection of real partial-label datasets. Code, frozen protocols, raw per-cell results with code and data hashes, and the scripts that generate every number, table and figure of this paper are available in an anonymised repository and will be released.

Licences. TabICL: BSD-3-Clause; TabDPT and the `tabpfn` package: Apache-2.0 (as declared in their package metadata). The TabPFN v2 and TabPFN-3 weights are distributed by their publisher under its own terms, which the user accepts before download for TabPFN-3; we used all checkpoints for research only and redistribute none of them. OpenML datasets are used under the licence listed on each dataset's page; the PALM datasets as distributed by their authors; we redistribute no dataset. Baseline code without a licence is loaded at run time and not redistributed.

Compute. All runs used free Kaggle sessions with two NVIDIA T4 GPUs (TFM-based methods) or CPUs (baselines). Study 1 used about 42 hours of two-GPU sessions and 26 hours of four-process CPU sessions; Study 2 about 4 GPU-hours and 80 CPU process-hours; the exploratory lots of Studies 1 and 2 about 13 hours of two-GPU sessions; Study 3 three two-GPU sessions of 8 to 11 hours; the development runs of Appendix G six further sessions of at most 11 hours; the supplementary analysis two two-GPU sessions (11 and 4.5 hours) and one CPU session of about one hour. Development and pilot runs before each freeze are not included.

Use of large language models. An AI assistant was used for writing code, drafting proofs that the authors checked, running analyses through the pre-written scripts, and drafting text. The authors designed the studies, approved every pre-registration and decision, ran all experiments and are responsible for all content.